BrainNet Viewer User Manual

MATLAB Desktop Edition

Version 2.0 development build

Mingrui Xia

National Key Laboratory of Cognitive Neuroscience and Learning
Beijing Key Laboratory of Brain Imaging and Connectomics
IDG/McGovern Institute for Brain Research
Beijing Normal University, Beijing, China

Contents

This manual explains how to load and validate brain visualization data, configure surface, node, edge, volume and ROI displays, inspect the result, save reusable Scenes and export figures. It also covers mesh tools, command-line rendering, migration and troubleshooting. Verify coordinate alignment, statistical meaning and resource permissions for each dataset before interpreting or sharing a figure.

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1 Installation and launch

BrainNet Viewer is a MATLAB desktop application for visualizing brain surfaces, networks and mapped volumes. This chapter explains installation requirements, startup and the main workflow.

Supported starting point

The primary tested desktop environment is **Windows with an installed, activated MATLAB R2025a Update 1** and a graphics-capable session. Focused app, layout, dialog and raster-export checks have also passed on macOS R2024a/R2026a, but a complete macOS package is not certified. MATLAB Runtime-only installation, Linux packaging and a standalone executable are not certified. The independent modern NIfTI/GIfTI readers do not require SPM, but individual optional features can need MATLAB toolboxes or GPU support. See the project README and the migration limits; a successful launch does not verify every optional feature.

Set MATLAB Current Folder to the installed software root, the directory containing the top-level BrainNetViewer.m, then run:

```
app = BrainNetViewer;
```

The launcher adds `src/matlab/BrainNetViewer` itself. From elsewhere, add only the installed software root to MATLAB's path first. Do not use `addpath(genpath(...))`, mix different software versions, or assume that a different installed `BrainNetViewer` is the intended installation. In MATLAB, `which BrainNetViewer -all` helps diagnose a path collision. No installer is supplied by this chapter.

What should appear

The English main window opens with its startup image/build information. Its three working areas support the intended sequence: **Data Sources** → **Visualization Settings** → **render in the main canvas** → **interact/export**. Use **Add Files** or role-specific slots, **Validate and Load**, change settings, then the shared **Apply**; Apply commits and renders. **Render Scene** is an explicit redraw.

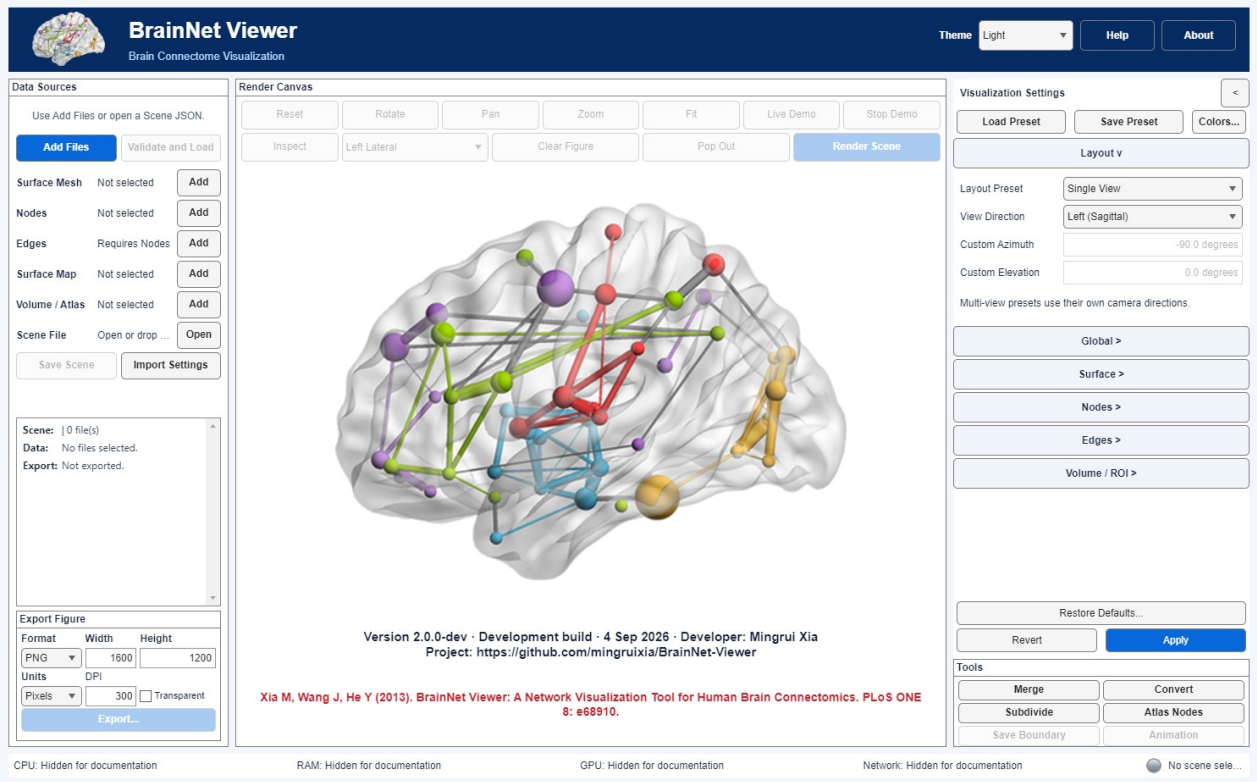


Figure 1. The English startup window identifies the data, render and settings areas before files are loaded.

Start with the real ICBM152 surface exercise, not an arbitrary template found in a data directory.
For file roles, coordinate cautions and data-root behavior, continue to Chapter 2. For saving an analysis, use Chapter 3.

Before sharing an image or package

The UI version/build date identifies this development build, not a public release date. A figure's provenance may contain local file paths. **Directory membership is not redistribution permission**; check source credits and package notices before sharing teaching data. The internal Beta 1 ZIP and the current workspace may differ. Review the changelog and the actual package contents before asserting version or compatibility claims.

2 Data formats and coordinates

Choose a file's **role**, not just its extension. In **Data Sources**, **Add Files** can classify candidates; use the individual slots when classification is ambiguous. **Validate and Load** reads and checks the selected combination before a figure is interpreted. The default desktop data root is `src/matlab/BrainNetViewer`, not the Scene JSON's folder.

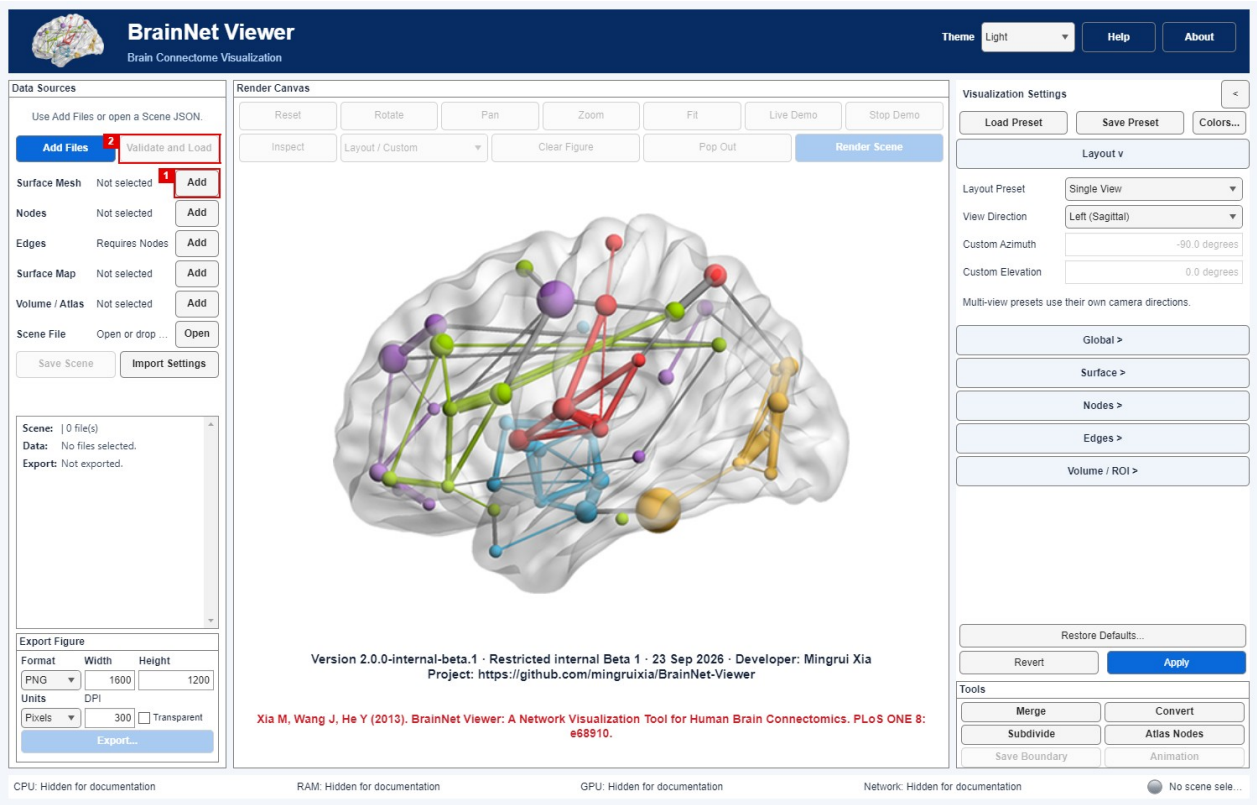


Figure 2. Step 1 selects Surface Mesh Add; step 2 validates and loads the selected inputs.

Role in Data Sources	Typical input	Key correspondence rule
Surface Mesh	`.nv`, `.gii/.gii.gz`, `.pial`, `.mesh`, `.g`, MNI/BIC `.obj`, `.mz3/.mz3.gz`	Geometry supplies XYZ and faces; a readable surface alone does not define MNI/subject space. See Chapter 13.
Nodes	BrainNet `.node`	Each row is a marker with coordinates and display fields. Preserve row order when pairing with an edge matrix; illustrative columns are not automatically measured quantities. See Chapter 7.
Edges	BrainNet `.edge`	Matrix rows/columns must correspond to node rows; directedness and threshold semantics depend on settings. See Chapter 8.
Surface Map	Original-order per-vertex scalar text	One value per mesh vertex in the **same vertex order**. Equal length does not prove atlas correspondence; see Chapter 6.
Volume	NIfTI-1/2 `.nii/.nii.gz` or supported paired `.hdr/.img`	Header/affine and statistic metadata matter. A readable volume need not be registered to a separate surface; see Chapter 9 and Chapter 10.
Scene File	Modern `.scene.json`	An analysis reference, not a mesh or volume; its files are resolved under the configured data root. See Chapter 3.

Choose a useful input combination

You do not need to load every role for every figure. A surface alone shows geometry; nodes alone show spatial markers; adding an edge matrix to nodes shows a network without an anatomical mesh. Combining a surface with nodes or with nodes and edges places those elements over anatomy. A scalar surface map or a volume-derived surface map requires a matching surface. The resulting map can also be shown with nodes and edges. To draw atlas or statistical ROIs, load a suitable volume and choose the ROI Volume Type; a surface makes their anatomical context easier to judge. An edge file without its matching nodes cannot establish endpoint locations.

The nine examples below illustrate the different input combinations. For an analysis, start with the fewest data roles needed and add one file at a time. After each addition, inspect the Scene Summary and check that the layers are spatially consistent. Colors and camera positions can be adjusted independently of the input data.

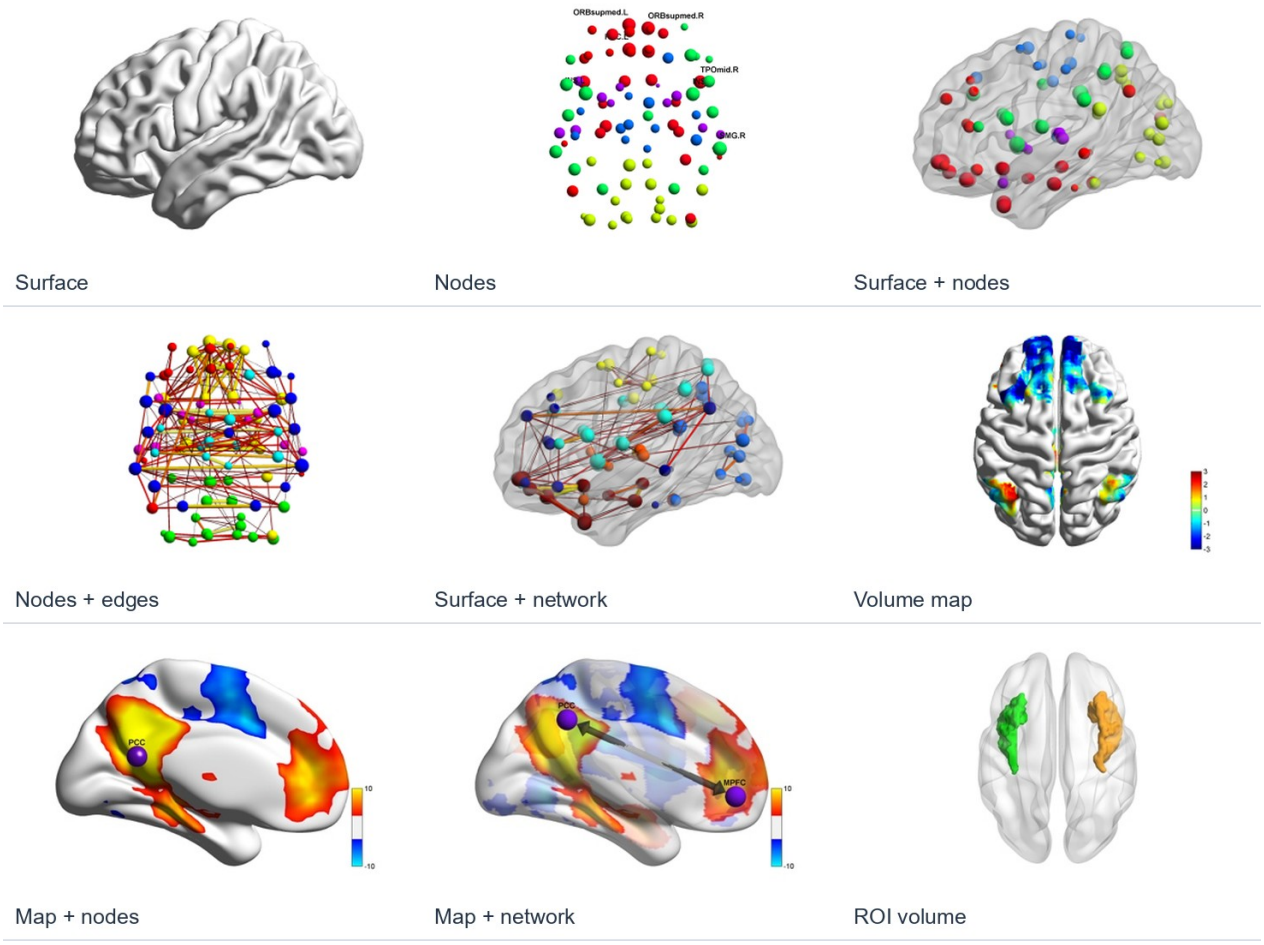


Figure 3. Data-role combinations for surface, node, edge and volume visualization.

Prepare text-based inputs

The BrainNet `.nv` format begins with an integer vertex count, then that many XYZ coordinate triplets, an integer triangle count, and that many triplets of **one-based** vertex indices. The number of coordinates and faces must agree with the declared counts. A `#` comment may document provenance, but it does not change coordinates or supply a registration transform. Mesh utilities in Chapter 13 can create a new `.nv` from supported formats after a review step.

A `.node` file has one node per line: `X Y Z color_or_module size label`. The first five fields must be finite numbers; the final label may contain spaces. A label consisting of `-` means no visible label. Column 4 may encode a continuous color value or a discrete module ID, depending on the Nodes color setting; column 5 controls mapped or raw radius. The following three rows are a **format example only**, not atlas coordinates:

```
0 0 0 1 2 Region_A
10 0 0 2 3 Region_B
0 10 0 2 1 Region_C
```

The matching `.edge` file is a whitespace-delimited 3×3 numeric matrix in precisely the same node order. For example, the first row below describes values from Region_A to Regions A, B and C:

```
0   0.4 0
0.4 0   -0.2
0   -0.2 0
```

Zero denotes an absent edge. In undirected mode, only eligible upper-triangle entries are drawn; in directed mode, ordered off-diagonal entries can be drawn separately. A file may contain `#` comments. Do not treat this illustrative matrix as a real functional-connectivity estimate. For a per-vertex text Surface Map, provide exactly one finite scalar per vertex in the mesh's **original vertex order**; a matching count without a trustworthy vertex correspondence is insufficient.

Atlas families are not interchangeable node orders or coordinate spaces. Check the package's file inventory, attribution and each sample's node count before pairing any .node with an .edge. Do not infer that a numerical atlas ID is a statistical effect size.

The independent modern NIfTI/GIfTI readers do **not** require SPM. Plain Analyze 7.5 `.hdr/.img` lacking NIfTI magic is not accepted. GIfTI external binary arrays are supported within the reader's validation rules; keep companions together when moving data. For surface formats, `.obj` means **MNI/BIC**, not Wavefront; `.mesh` means the supported BrainVISA encoding, not arbitrary Medit data. FreeSurfer `.pial` geometry is kept in stored surface coordinates; it is not automatically scanner-RAS or MNI. BrainVISA geometry is kept as stored rather than applying the old implicit XYZ inversion. GIfTI transforms/metadata may be recorded without applying registration. Format support does not imply that embedded MZ3 scalars/RGBA are automatically used for display. Consult the detailed matrix and Chapter 13 for explicit rejections.

Outside-root files and portability

For a file outside the App's configured root, the desktop asks **Stage and Add** or **Cancel**. Approval makes a verified copy under `user-data/imports/<hash>/` and stages supported companions, leaving the original untouched. It does **not** transfer rights to publish the copy. A programmatic `selectInputPath` requires explicit external-import approval; the CLI does not silently stage arbitrary absolute paths. A portable Scene keeps **relative references and hashes**, but not the bytes of the input files. Preserve their relative directory structure beneath the destination data root. See Chapter 3.

Scientific alignment check

Before combining inputs, identify species/template or subject, XYZ convention and handedness, physical units, surface vertex order, volume affine, and atlas/statistic meaning. Volume mapping may use the NIfTI affine to sample voxels, but **does not register two datasets**. Inspect assumes compatible MNI-like coordinates for its AAL/Brodmann lookup and does not perform an affine registration. A file hash establishes identity, not spatial alignment, license clearance or statistical validity. If any of these are unknown, stop anatomical interpretation and document the uncertainty. The real AAL vertex-map lesson and T-map lesson show exact-file examples with their limits.

3 Scenes and presets

Choose the right file

Need	Use	What it contains
Reopen a complete visualization	Scene JSON	Input references and hashes, applied visualization settings, and supported camera/export session state; not the data bytes
Reuse a look on different data	Visualization Preset	Display settings only; no input files, camera or export destination
Recover a trusted old project	Legacy configuration import	A migration starting point that requires field-by-field review, not a lossless modern Scene

Save and reopen a Scene

Load and validate the inputs, configure the scene, and select **Apply** to commit and render. Save the Scene JSON from the main window. The Scene records paths relative to its configured data root where possible and verifies referenced inputs by hash. It does not embed mesh, node, edge or volume data. Keep the inputs and their compatible directory structure with the JSON when moving a project. If files are outside the configured root, follow the explicit import/staging workflow in Chapter 1; selecting an external file does not silently grant redistribution rights.

When reopening, review resolved paths, hashes, file roles, spatial alignment and the rendered result. A missing or changed input is a reason to investigate, not to bypass validation. Scene replay can restore supported camera and export preferences, but an old Scene may not express every newer option. Scene Workflows documents the detailed schema, path and command-line behavior.

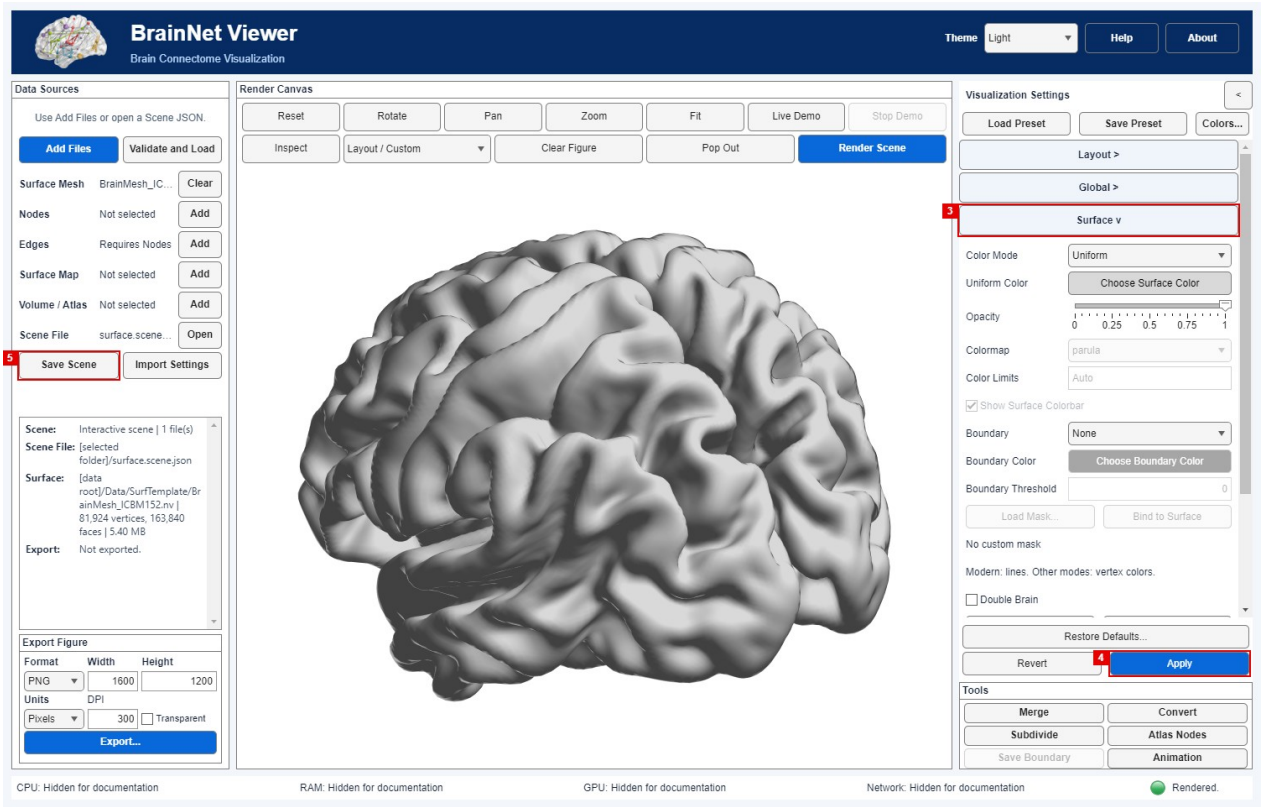


Figure 4. A saved Scene reopened with an opaque gray surface. Step 3 opens Surface settings, step 4 applies and renders, and step 5 saves a Scene.

Apply a visualization preset

Choose a preset after loading the intended data. Preset load replaces the visualization settings sections; it does not bring in files or preserve pending edits and camera state. Review the resulting controls, then use **Render Scene** for a visible redraw. Save a new Scene JSON if the input-plus-display combination should be reproducible. See Visualization Presets for the supported fields and reset behavior.

Import a legacy configuration

Use a legacy MAT configuration only when its source is trusted. The migration translates what it can, reports fields requiring review, and may change the rendered appearance. Inspect file roles, paths, layout, thresholds, colors and export options; compare a representative image before adopting the result. Review the translated settings before use. Migration and troubleshooting lists current limits.

Batch and provenance

For repeatable non-interactive rendering, use `BrainNetRender` with a reviewed Scene. The Scene JSON, input checks and export provenance help reconstruct a figure but do not replace recording data origin, atlas version, coordinate convention, software build and any manual decisions. Chapter 14 covers batch use. Before sharing any Scene, inspect paths, metadata and sidecars for private local information; sharing the JSON never authorizes sharing its referenced data.

4 Layout and camera controls

Use **Visualization Settings** → **Layout** to choose a view arrangement and initial camera direction. The shared **Apply** commits changes and renders; **Revert** discards pending edits. Saved Scenes and presets retain applied layout settings, but presets do not carry the current camera.

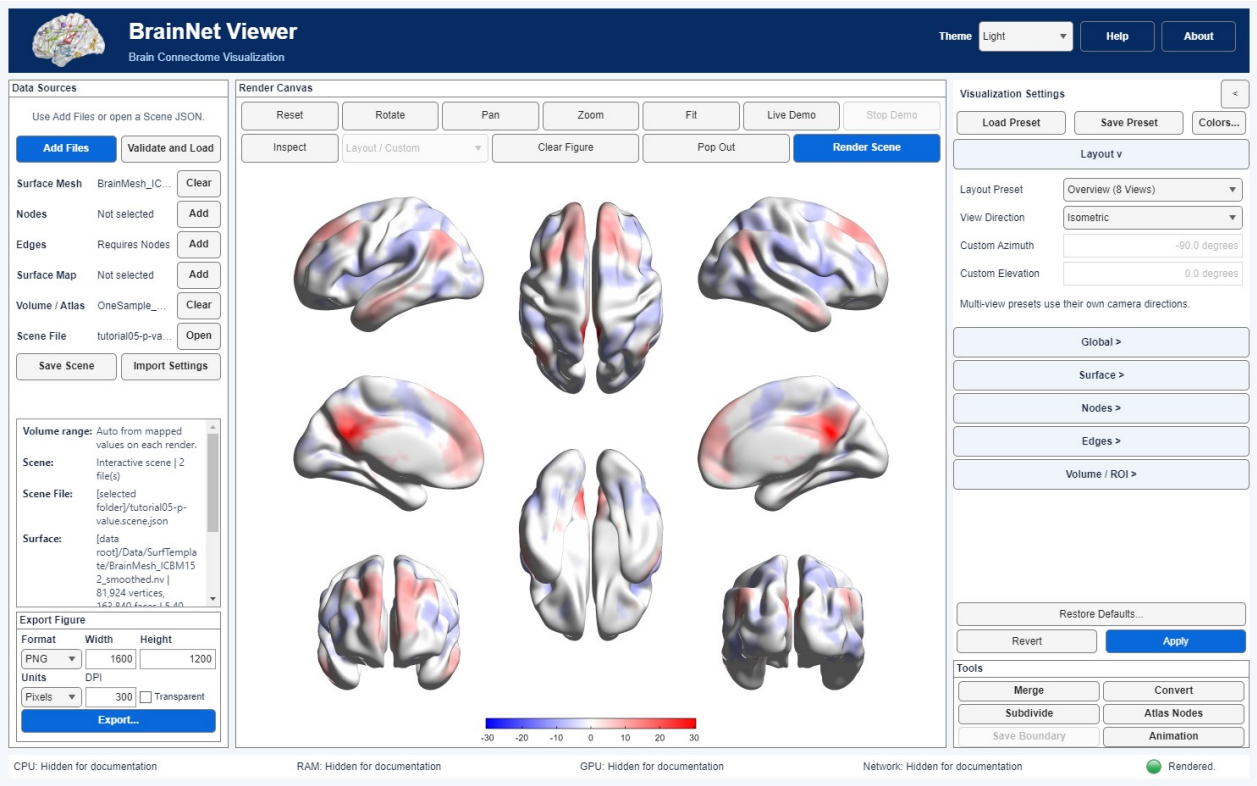


Figure 5. Overview with eight views for a surface that can be split into hemispheres.

Control	New-settings default	Effect and when to use it
Layout Preset	**Single View**	Five choices: Single View; Overview (6 / 8 Views); Lateral / Medial; Lateral / Medial / Ventral; Lateral / Medial / Dorsal. Multi-view presets set their own pane directions and use a shared scale; they are not independent copies of the single-view direction control.
View Direction	**Left (Sagittal)**	Single-view directions: Left, Superior, Anterior, Custom, Right, Inferior, Posterior, Isometric. In multi-view, the preset's own directions take precedence. A named view is a camera orientation, not proof of anatomical L/R.
Custom Azimuth	no explicit value	Enabled for Custom direction; numeric degrees, −360 to 360. Save and replay the applied value in Scene/preset.
Custom Elevation	no explicit value	Enabled for Custom direction; numeric degrees, −90 to 90.

Read a multi-view figure

Single View is best for adjusting one camera or inspecting one selected point. The lateral/medial arrangement places the outside and inside of each separable hemisphere in four panes. The ventral variant adds inferior views; the dorsal variant adds a superior overview. **Overview (6 / 8 Views)** provides a broader orientation survey: a safely separable two-hemisphere mesh can use eight panes, while a single or inseparable surface uses six complete-data views. Choosing a multi-view preset is not the same as loading multiple independent datasets: every pane represents the same Scene from another camera.

The layout gallery illustrates the available viewing arrangements. Verify the dataset coordinate convention before interpreting orientation labels. At the intended export size, check that labels and thin edges remain legible in the smaller panes.

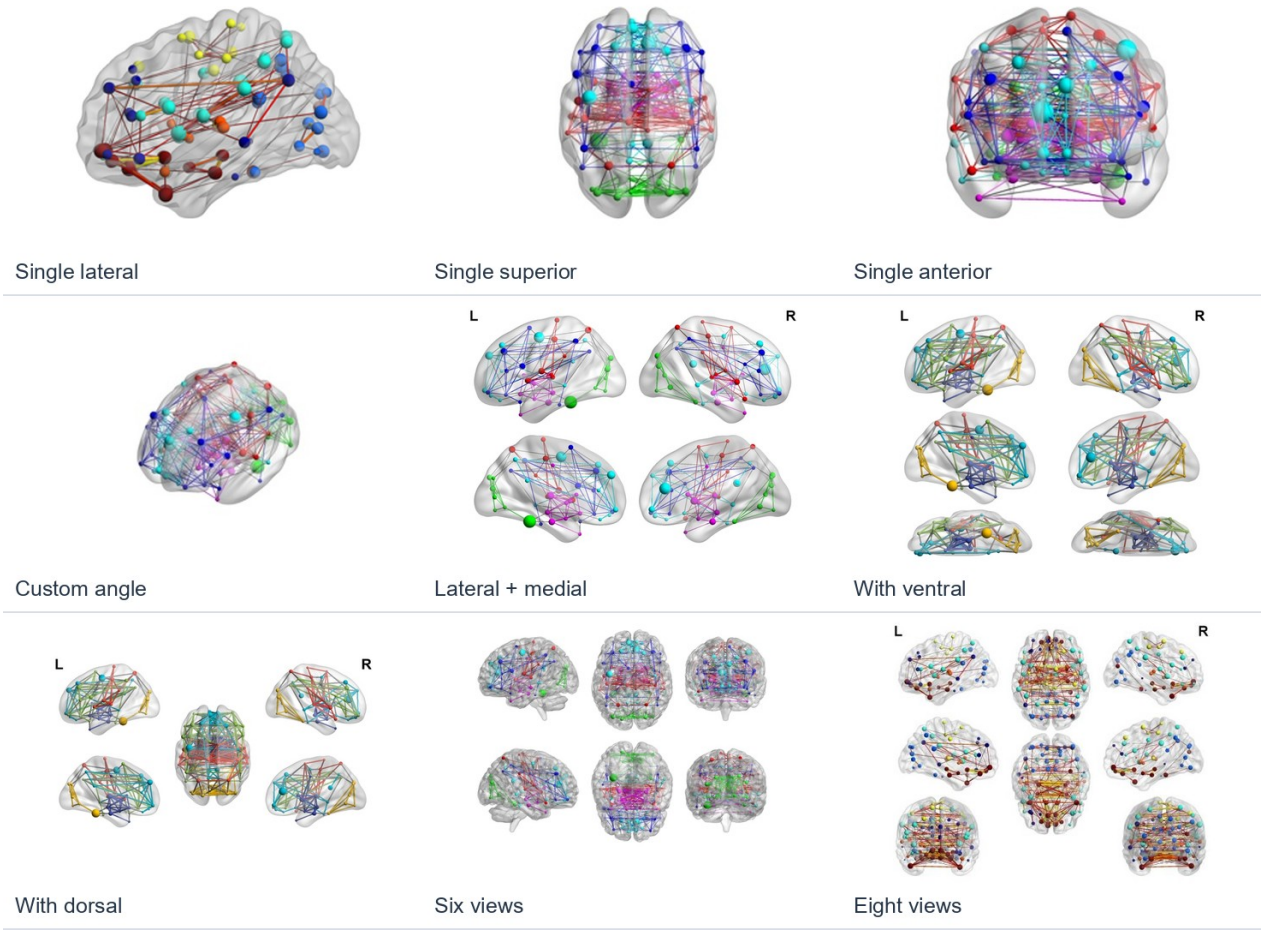


Figure 6. Viewing arrangements and relative positions of the brain panes.

Overview (6 / 8 Views) uses eight split views only when a surface can be safely partitioned into nonempty, vertex-disjoint X-side groups. Otherwise it shows six complete-data views and does not invent a missing hemisphere. This conservative rule does not segment arbitrary anatomy or verify registration. A saved eight-view camera is reset with a notice if replay resolves to six views, and conversely; see Layout 2.

After rendering, use the canvas controls to **Rotate**, **Pan**, **Zoom**, **Fit**, or **Reset**. Fit recenters and frames visible geometry without changing input coordinates; Reset restores the selected layout's initial camera. Resizing preserves the current camera rather than silently fitting it. **Pop Out** opens an independent inspection copy; later movement there does not replace the main canvas camera. **Save Scene** records a valid modern main-canvas camera; **Save Preset** does not. See camera/replay details and window sizing.

Double Brain (described in Chapter 6) is a separate single-view display option, not a sixth layout. It is restricted to Left, Superior or Anterior direction with a surface and no scalar-map/volume input. The pane badges governed by **Show Orientation Labels** are described in Chapter 5; they identify coordinate halves unless an input-bound anatomical declaration has been explicitly verified.

Multi-view readability depends on mesh shape, labels and display scaling. Check the final figure at its intended export size.

5 Global appearance and rendering

Use **Visualization Settings** → **Global** for the canvas background, shading, lighting and rendering detail. **Apply** commits and renders. A Scene or preset may contain different explicit values; the table describes new settings, not a promise to overwrite loaded configurations.

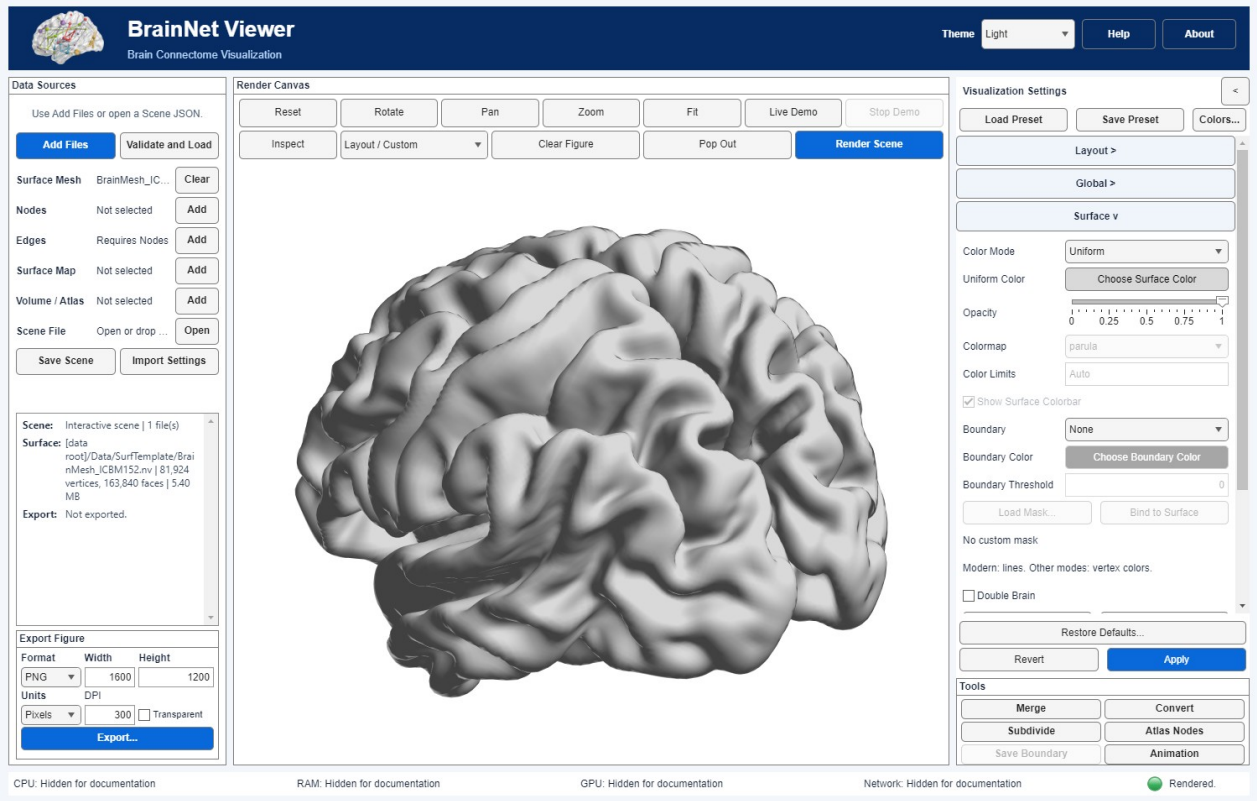


Figure 7. An opaque gray ICBM152 surface with Surface settings visible.

Control	New-settings default	Meaning / constraints
Background Color → Choose Color	white `[1 1 1]`	Sets the scientific canvas background, not the Light/Dark interface theme. App export set to **current** follows the rendered canvas color.
Surface Shading	**Interpolated**	Flat, Faceted or Interpolated surface face/vertex shading. Faceted draws black mesh outlines and may cost more on dense meshes. This is display shading, not mesh smoothing.
Lighting Model	**Phong**	None, Flat, Gouraud or Phong lighting for shaded geometry. Appearance depends on material, view and graphics backend.
Light Direction	**Right**	Headlight, Right or Left. Camera/light behavior should be inspected after manual rotation; the choice does not change data coordinates.
Surface Material	**Dull**	Dull, Shiny, Metal or Custom. Custom enables the three strength fields below; presets follow MATLAB's material behavior.
Ambient / Diffuse / Specular Strength	`[0.5 0.5 0.5]` stored	Each accepts 0–1 and is editable for Custom material. Values affect lighting response, not underlying scalar data.

Control	New-settings default	Meaning / constraints
Render Detail	**Adaptive (modern)**	Adaptive uses load-aware renderer policy. Legacy Low/Moderate/High use node sphere detail 10/50/100 and edge-cylinder sides 5/10/20; High can be costly. The surface topology is unchanged.
Show Orientation Labels	off	Multi-view pane badges are X<=0 / X>0 unless the Scene has an explicit, input-bound anatomical-X declaration; only then can they say L/R. They label panes, not the screen's left/right edge. Single View and six complete-data Overview have no half-space badges.

Compare appearance settings

Material changes the balance of ambient, diffuse and specular reflections. **Dull** gives a relatively subdued shape; **Shiny** and **Metal** emphasize highlights. A Custom material lets you adjust the three strengths rather than replacing the input mesh. Surface Shading controls whether neighboring vertex colors are interpolated and whether triangle boundaries are visible; use Faceted to inspect mesh edges, not to infer an anatomical boundary. The Lighting Model controls how illumination is calculated, while Light Direction chooses where it comes from. These are separate decisions: an interpolated color map can still look flat under weak lighting, and changing illumination never changes a statistic in the source map.

The appearance gallery compares material, shading and light direction on one geometry. Set the palette and statistical range first, choose lighting and material, then check legibility at the intended export size. Use Render Detail conservatively on dense networks: higher sphere and cylinder tessellation increases geometry without adding measurements or changing surface topology.

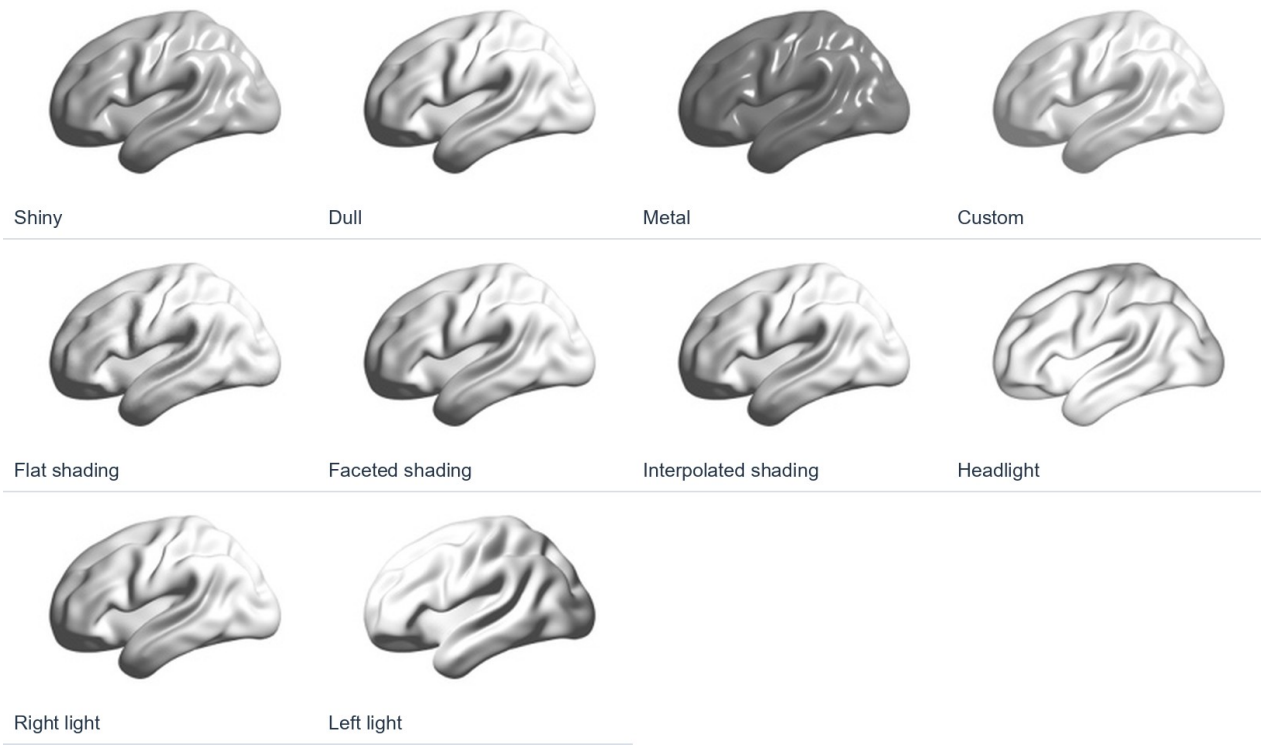


Figure 8. Surface material, shading and light-direction examples.

Theme in the top bar changes UI colors only. For export, **current** preserves the actual scientific canvas background; transparent raster requests may be rejected rather than silently flattened. See backgrounds. Lighting/material specifics and the exact legacy geometry table are in Material Settings and Render Detail.

The App does not infer anatomy from a filename or species. To show anatomical L/R, first verify coordinate units, shared registration, the X=0 midline and positive-X direction, then attach the declaration to hashed Scene inputs using the documented Scene/API workflow. Merely ticking **Show Orientation Labels** without that declaration shows coordinate halves. Check label contrast on the display and in the exported figure.

Restore Defaults resets display settings while keeping the current canvas visible. Click Apply or Render Scene to redraw; export is paused until the retained preview is replaced. It does not return to the welcome page.

6 Surface appearance and maps

Use **Visualization Settings** → **Surface** to set the surface color, opacity, boundaries and vertex colors. **Apply** commits and renders. A loaded Scene or preset keeps its explicit values unless you edit them.

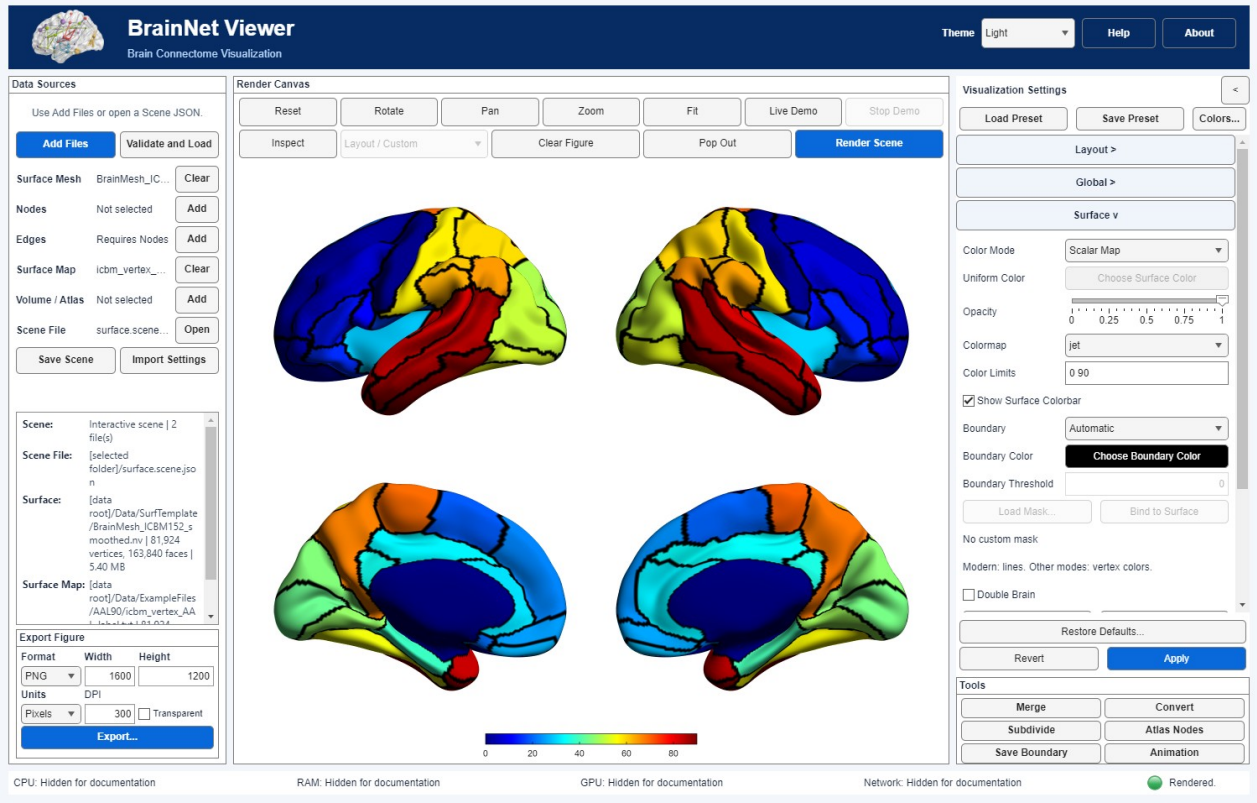


Figure 9. AAL vertex labels with automatic boundaries in four lateral and medial views.

Control	New-settings default	Meaning / enabling rule
Color Mode	Uniform	**Uniform** uses one RGB surface color; **Scalar Map** colors vertices from a loaded surface map or supported volume-to-surface mapping; **Vertex RGB** uses an explicitly bound original-order RGB table. Choose the mode that matches the data, not merely a desired palette.
Uniform Color	`[0.95 0.95 0.95]`	Active only for Uniform; **Choose Surface Color** changes the single color.
Opacity	Settings default 0.3	Slider range 0 (transparent) to 1 (opaque). A new surface-only or surface-plus-map/volume Scene instead starts at **1**; explicit Scene/preset values are preserved. This display opacity does not change input values.
Colormap	Parula	Active for a loaded scalar surface map. Choose a named palette or **Custom (Colors...)**. Volume-to-surface coloring uses the **Volume** panel's separate colormap; Vertex RGB bypasses scalar palettes.
Color Limits	Auto	For Scalar Map, enter `Auto` or two numeric endpoints (for example `-2 2`). Fixed limits control color scaling, not vertex-value filtering. Recheck them when switching datasets.

Control	New-settings default	Meaning / enabling rule
Show Surface Colorbar	on	Shows one continuous colorbar when a scalar surface map or volume-to-surface map is displayed; Uniform and Vertex RGB have no scalar bar. Existing Scenes/presets retain an explicit off setting. Atlas ROI ID legends are separate categorical keys.
Boundary	None	**Automatic** draws class-transition lines; **Threshold** marks vertices at a strict value threshold; **Custom Mask** uses an original-order binary mask. Boundary operations need the relevant map except Custom Mask.
Boundary Color	black `[0 0 0]`	Active when a boundary mode is selected; changes display color only.
Boundary Threshold	0	Numeric field used only by Threshold; its classification is `value > threshold`, not `>=`.
Load Mask... / Bind to Surface	no mask	Available for Custom Mask. Import or bind a binary table only after verifying original vertex order; a matching count alone cannot prove anatomical correspondence. The status line reports binding state.
Double Brain	off	Displays a second copy of one surface in Single View, Left/Superior/Anterior only; optional node/edge data follow explicit row ordering. No surface-map/volume input. This is not arbitrary two-subject composition.
Load RGB... / Bind RGB	no RGB table	Text is N×3 finite RGB in [0,1] for one brain, or 2N×3 for Double Brain, in original surface vertex order. Binding is required; it does not independently verify initial correspondence. The status line reports binding state.

Double Brain and boundary examples

Double Brain copies **one** surface for a side-by-side network comparison. If a node file contains N rows, rows 1 through `floor(N/2)` belong to the first display and the remaining rows to the second. For example, a 180-row file may place 90 matched regions in each copy. Its edge file is then 180×180: the upper-left and lower-right 90×90 blocks describe within-copy connections, and the two off-diagonal blocks describe connections between copies. The program does not discover which rows correspond to the same regions or verify a longitudinal, group or subject interpretation. Prepare and review the ordering before turning on Double Brain; an ordinary 90×90 edge matrix is not the paired network for a 180-row node file.

The boundary gallery compares Double Brain, categorical atlas outlines and continuous mapped color. A region outline is a display boundary, not a new ROI volume. Continuous colors require a scale and must not be interpreted as category IDs. Automatic, Threshold and Custom Mask boundaries follow the rules above. Recheck visibility when changing surface opacity, map range or export resolution.

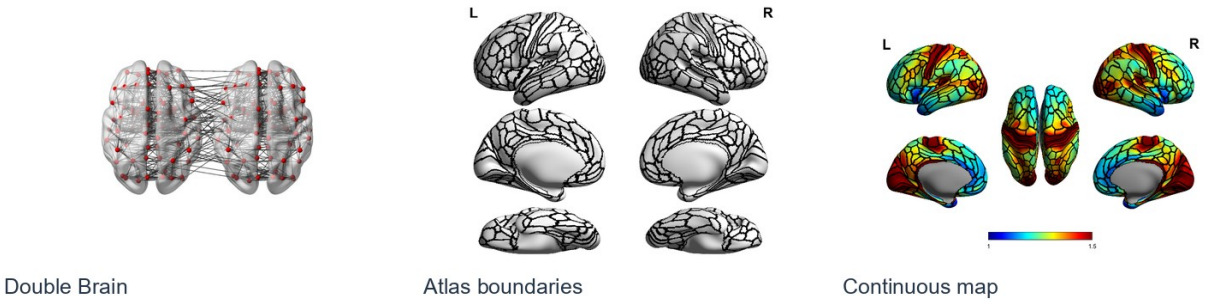


Figure 10. Double Brain, atlas boundaries and continuous surface coloring.

Automatic in the current panel is the modern line-style boundary option; an older Legacy Automatic setting can be read for compatibility but is not a separate current panel choice. Threshold and Custom Mask use vertex colors rather than geometric isolines. See boundary algorithms and mask safety. **Vertex RGB** preserves the original mesh while active and can cost more for large surfaces; it is not a scalar map or statistical threshold. See RGB format/binding and Double Brain ordering.

The **Volume** section controls volume mapping algorithm, sign/threshold handling and ROI drawing; these are separate from Surface's scalar palette and are described in Chapter 9. A visually plausible map is not evidence that the mesh and volume are registered. Confirm coordinate spaces before anatomical interpretation.

For an AAL vertex-label exercise with Automatic boundaries and binary-mask export, see tutorial 04. Atlas IDs are categories, not effect sizes. Use the matching smoothed mesh and retain the resource credits.

Inspect on the rendered canvas

Click Inspect, then pick a displayed surface vertex or node. The on-canvas label reports source index, available value/module, assumed-MNI XYZ, Brodmann Area and AAL region. Another pick replaces it; clicking Inspect again clears it. Edges, arrows, ROI meshes and fibers are not pick targets. Surface picks use the nearest displayed source vertex rather than interpolating the mouse-hit position.

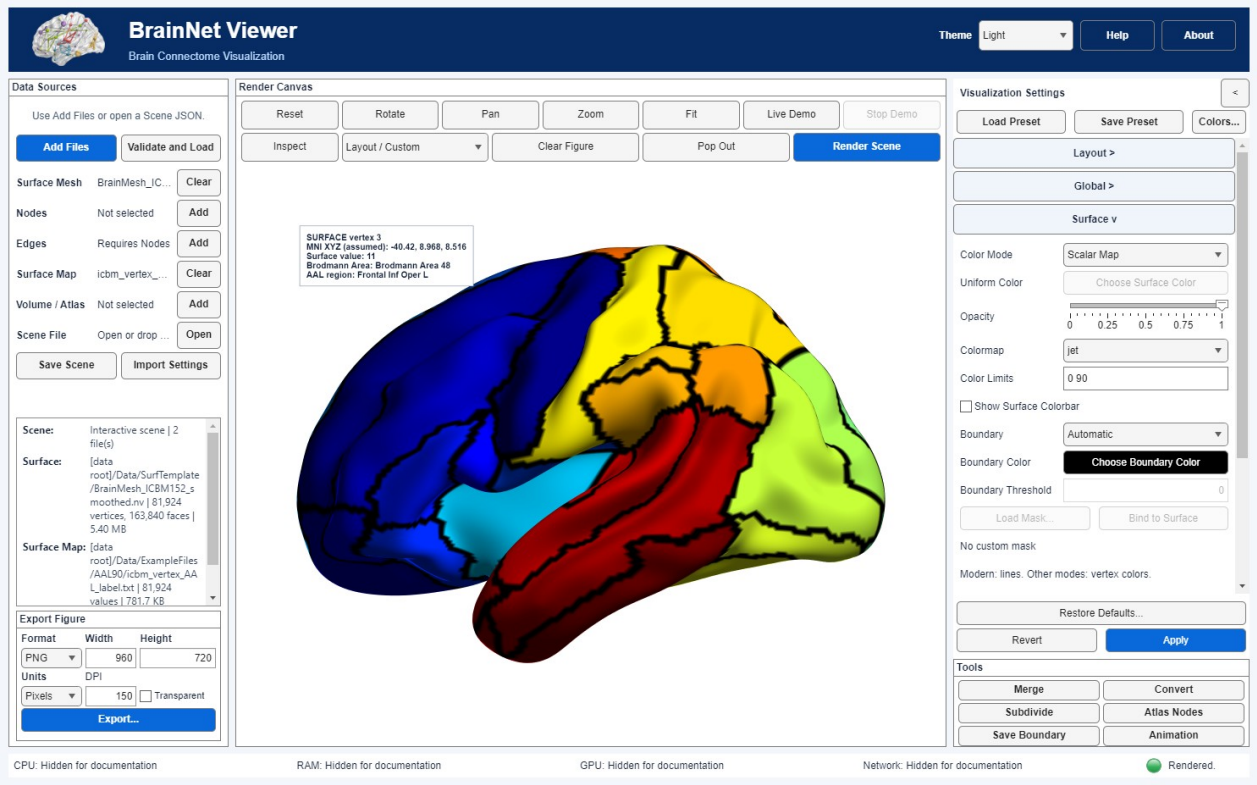


Figure 11. An Inspect label on the rendered surface reports Brodmann and AAL regions.

Inspect assumes coordinates are already in the relevant MNI space. It does not register data or apply transforms. Native-space, animal or unregistered inputs may give plausible but incorrect labels. Verify coordinates before anatomical interpretation; see Inspect limits.

The AAL vertex-map exercise demonstrates this lookup with the bundled AAL/Brodmann resources. A successful lookup does not prove spatial registration.

7 Nodes

Use **Visualization Settings** → **Nodes** to control marker visibility, labels, size and color. In a `.node` row, columns 1–3 are XYZ coordinates, **column 4 supplies module/color values**, column 5 supplies node-size values, and column 6 is a label. Verify units and anatomical space rather than inferring them from the extension. The shared **Apply** commits and renders; **Revert** discards pending edits.

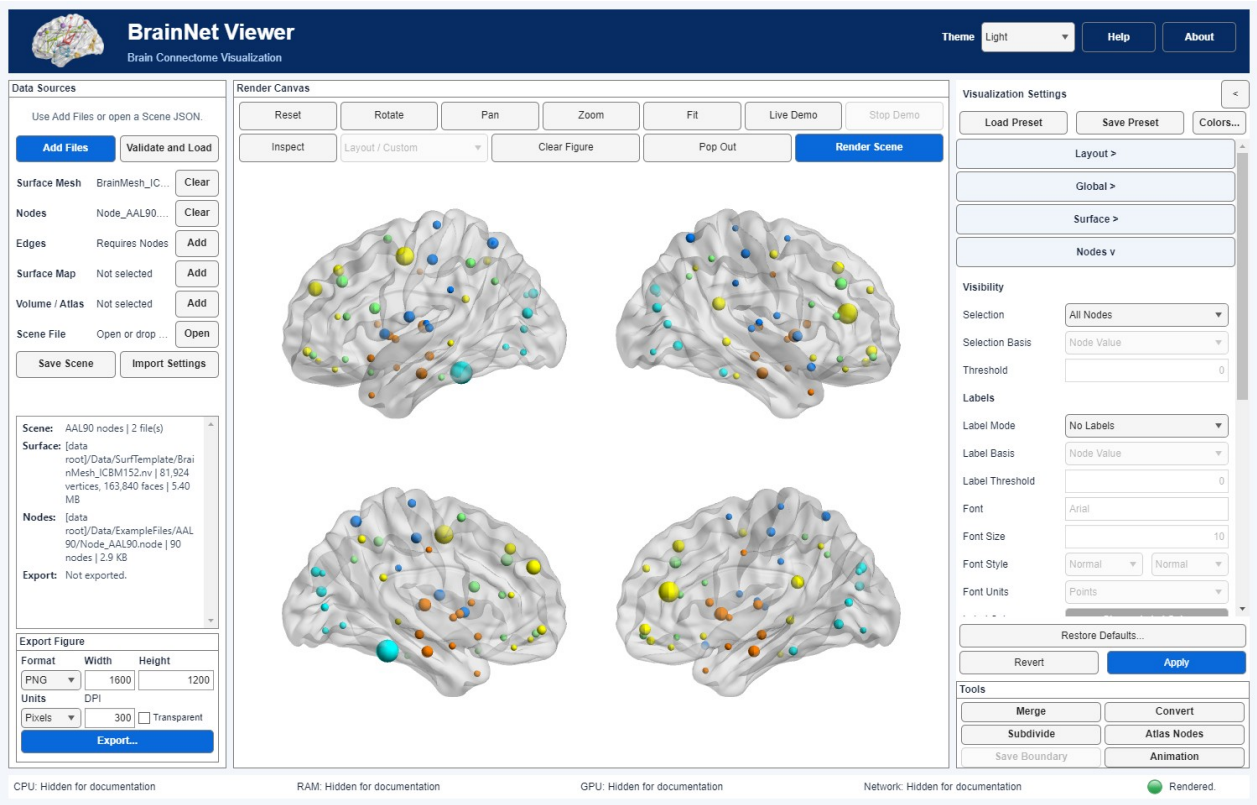


Figure 12. The bundled AAL90 nodes on ICBM152 in four views; colors use node column 4 and mapped sizes use column 5.

Visibility and labels

Control	New-settings default	Current rule
Selection	All Nodes	Threshold keeps nodes with original **Selection Basis > Threshold**; equality is excluded. Connected Nodes keeps either endpoint of an edge retained after edge filtering; no selected edges means no Connected nodes.
Selection Basis / Threshold	Node Value / 0	Enabled for Threshold. Node Value reads column 5; Module reads column 4. Module still uses a numeric > comparison, not equality to a chosen ID. Mapped radii and Scale do not affect this test.
Label Mode	No Labels	All Labels, No Labels, or Threshold. Labels appear only on visible nodes with nonzero rendered radius; a label exactly `-` is suppressed, but a hyphenated name remains valid.
Label Basis / Label Threshold	Node Value / 0	Enabled for Threshold labels. Uses the original column 5 or 4 value with strict >; independent of node-selection threshold.
Font / Font Size	Arial / 10	Font name and a positive numeric size (editor limit >0 to 100). Font Units defaults to Points; other offered units are Pixels, Normalized, Inches and Centimeters.

Control	New-settings default	Current rule
Font Style / Label Color / Keep Labels in Front	Normal + Normal / black / off	Weight is Normal or Bold; angle Normal or Italic. The front option changes display order, not data coordinates; it cannot guarantee collision-free dense labels.

Node size

Control	New-settings default	Current rule
Size Mode	Mapped	Uniform uses Base Size; Mapped normalizes source values to radii; Raw uses unnormalized source values; Threshold uses either Mapped or Raw radius rules then replaces radii **strictly below** Size Threshold with 1. Equality remains.
Size Source	Node Value	Column 5 by default, or Module/column 4 as a modern extension. Inactive in Uniform. This choice does not alter color mode.
Base Size / Scale	3 / 1	Base Size matters in Uniform; positive Scale multiplies the resulting radius exactly once in all modes. Radius is in input-coordinate units, not screen pixels.
Size Threshold: Mapped/Raw, value	Mapped / 0	Enabled only for Threshold. Comparison occurs before Scale; a below-threshold radius becomes 1 before scaling. This threshold changes size, not visibility.

Mapped sizes follow the legacy normalization (including its log compression for large ranges); constant input gives radius 4 before Scale. Raw negative/invalid final radii are rejected rather than silently made positive. A zero final radius omits that sphere and its label but does not delete the input node or its edges. See numeric size rules.

Node color

Control	New-settings default	Current rule
Color Mode	Uniform	Uniform uses one Color; Mapped samples a palette from **column 4**; Modules assigns colors by sorted unique column-4 module IDs; Threshold uses two colors split by column 4. Column 5 is not the default color source.
Color / Colormap / Color Limits	RGB `[37 89 152]/255` / Jet / Auto	Color button is active in Uniform. Palette is active for Mapped or Modules and can be custom through Colors...; fixed `low high` limits apply to Mapped, clamping outside values. Auto uses the full input column-4 range, not just visible nodes.
Color Threshold / Low (<=) / High (>)	0 / `[37 89 152]/255` / red	In Threshold, column-4 values <= threshold use Low, values > use High. Mapped limits/palette do not change these two fixed colors.

For Modules, the palette comes from the **Nodes** colormap; same-module edge coloring can inherit it when Edges → Node Module is selected. A custom palette is not the same as the explicit, geometry-bound Vertex RGB surface table. See node color semantics, selection/labels and label placement.

Dense labels can overlap even when **Keep Labels in Front** is enabled. Inspect their placement at the intended export size and simplify labels or the view when necessary.

8 Edges

Use **Visualization Settings** → **Edges** to control connectivity selection, direction, size, color and opacity. An `.edge` matrix must follow the loaded `.node` row order. Values, coordinate distances and directed relationships have meaning only in that input's declared scientific context; neither edge lengths nor arrowheads imply white-matter tracts. The shared **Apply** commits and renders; **Revert** discards pending edits.

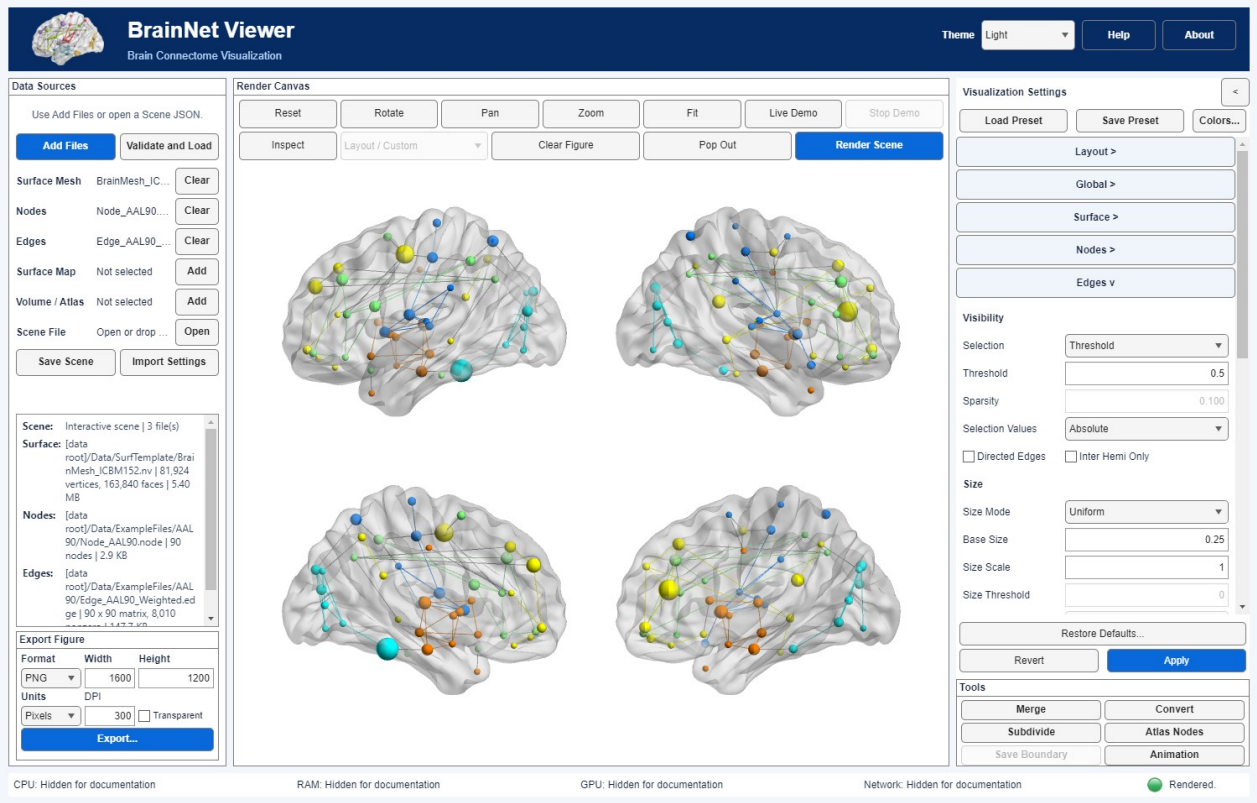


Figure 13. The bundled AAL90 weighted network in four views, showing undirected connections at threshold 0.5.

Visibility and direction

Control	New-settings default	Current rule
Selection	All Edges	All keeps nonzero eligible off-diagonal edges. Threshold keeps selection values `>= Threshold` (equality included). Sparsity (Top Count) ranks nonzero candidates and keeps a deterministic count. Sparsity (Legacy Threshold) ranks the full N×N matrix, converts its ranked value to a six-decimal cutoff, then keeps all eligible cutoff ties. These modes are **not** interchangeable.
Threshold / Sparsity	0 / 0.1	Threshold field is used only in Threshold; Sparsity fraction is 0–1 and used in either sparsity mode. Top Count fraction 0 selects none; Legacy Threshold fraction 0 still uses the largest matrix value and may show none if that value is diagonal.
Selection Values	Absolute	Signed or Absolute changes ranking/thresholding only; source weights are not rewritten. Other size/color/opacity signs have separate controls.
Directed Edges	off	Off uses nonzero upper-triangle candidates; on uses nonzero off-diagonal ordered pairs and draws arrowheads. The matrix is not automatically symmetrized.

Control	New-settings default	Current rule
Inter Hemi Only	off	After selection, retains opposite-X-side or midline endpoints by node X=0. This is a coordinate rule, not anatomical hemisphere detection.

Top Count resolves equal-weight candidates by MATLAB column-major linear index; Legacy Threshold can retain all ties, so final visible density need not equal the fraction. Connected Nodes in Chapter 7 uses the finally selected edge pairs. See exact sparsity and tie rules and X-side filtering.

Size and arrowheads

Control	New-settings default	Current rule
Size Mode	Mapped	Uniform uses Base Size; Mapped maps retained weights to radii 0.4–6 (constant set: 2); Raw clamps to 0.2–6; Threshold first tests original sign-adjusted weights `>= Size Threshold`, then sizes qualifying edges by Mapped or Raw basis. Below threshold uses radius 0.4, not removal.
Base Size / Size Scale	1 / 1	Base Size is used in Uniform. Positive Size Scale multiplies radii once in all modes. These are radii in input-coordinate units, not pixel widths.
Size Threshold / Threshold Basis	0 / Mapped	Enabled only in Threshold. Basis is Mapped or Raw; the threshold comparison precedes normalization and scaling.
Size Values	Absolute	Signed or Absolute is independent of Selection Values and color/opacity signs. Negative invalid final radii are rejected rather than silently made positive.
Arrowhead Shape: Length / Width	1 / 0.7	Enabled for Directed Edges. Multipliers are 0.25–3; length is capped at 35% of edge length. The edge tube ends at the arrowhead base to avoid overlap. These settings change geometry, not matrix direction or weight.

Threshold+Raw has a different lower-radius rule from ordinary Raw; see full radius calculation. A zero-radius edge may be absent from geometry without changing its selection/Connected semantics.

Color and opacity

Control	New-settings default	Current rule
Color Mode	Uniform	Uniform uses one Color. Mapped uses a 64-entry palette. Threshold uses Low when value <= Color Threshold and High when >; Length uses Short when original-node Euclidean distance < Length Threshold and Long at >=. Node Module compares both endpoints; Custom Matrix uses an explicitly bound node-order color-index matrix.
Color / Colormap / Color Limits	`[1 0.9 0.73]` / Jet / Auto	The Color button is for Uniform. Mapped/Custom Matrix use the selected named colormap or Custom (Colors...); fixed `low high` limits apply only to Mapped. Auto uses retained color weights after selected sign handling. Equal limits use the last palette color.
Color Threshold / Length Threshold	0 / 0	Threshold coloring uses original edge weights with its own sign choice; equality is Low. Length uses input XYZ coordinate units, not tract/geodesic length; equality is Long.
Threshold Values / Mapped Values	Absolute / Absolute	Separate Signed/Absolute choices for their respective color modes. They do not change Selection Values, radii or opacity.
Low / High colors; Inter-module Color	gray `[.38 .38 .38]` / `[1 .9 .73]`; gray `[.5 .5 .5]`	Low/High are used for Threshold or Length (buttons become Short/Long). Node Module uses the Nodes module palette for same-module edges and Inter-module Color for different modules. Select Nodes → Modules to make visible node colors match.
Load Colors... / Bind to Nodes	no matrix	Custom Matrix only. Confirm matrix dimensions and original node order; the binding cannot independently establish scientific correspondence.
Opacity Mode / Value / Range	Uniform / 0.7 / `[0.1 0.9]`	Uniform uses one 0–1 opacity. Mapped maps selected weights to two **output** opacity endpoints in [0,1]; constant weights use the upper endpoint. A transparent edge is not deselected.
Opacity Values	Absolute	Signed/Absolute for Mapped opacity only, independent of selection, size and color.

The **Node Module** palette is derived from the Nodes colormap; the Edges colormap selector does not override it. This mode supports at most 20 distinct modules and rejects excess categories rather than recycling colors. Custom Matrix colors require explicit binding and file-format review. Color threshold/length endpoints are detailed in Edge Colors; mapped opacity in Edge Opacity. Changing a visual option does not rewrite the `.edge` file or establish registration.

Before publishing, check dense or directed networks at the intended export size; an edge that is selected may still be hard to see because of its radius, opacity or overlap. The modern renderer is not guaranteed to match historical pixels exactly.

9 Volume mapping and statistics

Use **Visualization Settings** → **Volume** to map a 3-D NIfTI onto a surface. A readable NIfTI is not automatically aligned with a separately loaded surface. Mapping transforms surface coordinates by the volume affine to sample voxels; it does **not** perform image registration. Verify the surface/volume coordinate spaces and units before interpreting locations. **Apply** commits settings and renders; **Revert** discards pending changes.

In the Volume panel, **Threshold Mode**, **Statistical Threshold**, **P Value**, **Cluster Size** and **Voxel Connectivity** appear above the **Surface Mapping** section because they also govern statistical ROIs. Controls enable according to Volume Type and Threshold Mode; ROI P Value uses Cluster Size, while Surface Map P Value does not.

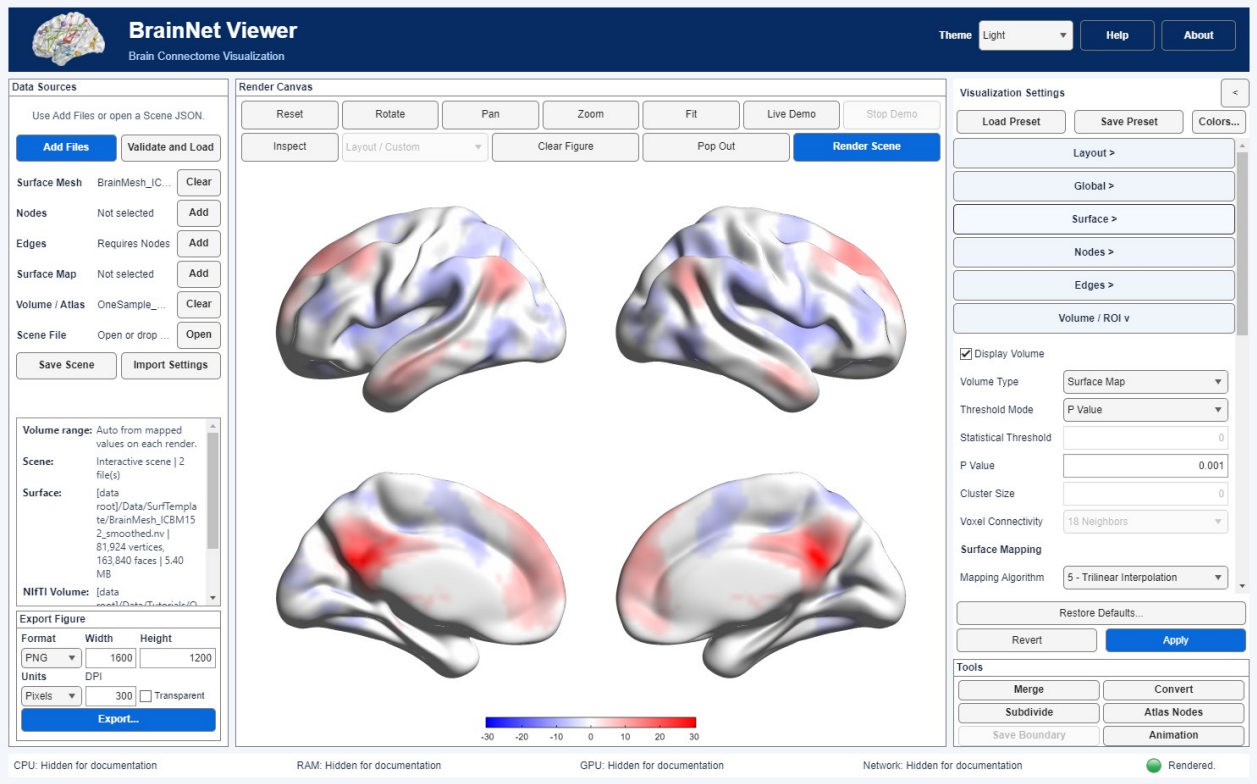


Figure 14. A signed statistical map displayed on four surface views with the P-value controls visible.

Control	New-settings default	Current rule
Display Volume	off	Enables the chosen Volume Type. A newly loaded file does not by itself establish scientific suitability.
Volume Type	Surface Map	Surface Map maps a 3-D volume onto a loaded surface; **ROIs (Atlas / Statistical)** extracts labelled meshes and uses the Chapter 10 controls.
Mapping Algorithm	2	Used for Surface Map only. The eight algorithms below have different numerical meanings, not a simple low/high-quality order.
Threshold Mode	None	None keeps values; Value keeps finite values with `abs(value) >= abs(Statistical Threshold)`; Cluster adds a minimum connected-component size. P Value converts supported NIfTI statistical metadata into a voxelwise cutoff. In ROI mode it also applies Cluster Size; see Chapter 10.

Control	New-settings default	Current rule
Statistical Threshold / P Value / Cluster Size / Voxel Connectivity	0 / 0.05 / 0 / 18 neighbors	Value and Cluster use the numeric threshold; Cluster also uses voxel count and 6/18/26 connectivity. P Value requires `0 < P < 1` and valid statistic metadata. The original volume is not rewritten.

For Value/Cluster, zero is background, nonfinite voxels are excluded, equality at the absolute-value threshold is retained, and Cluster removes components **smaller** than Cluster Size. The 6/18/26 selection defines statistical connectivity; it is not the separate 26-connected safety check used in ROI smoothing. Cluster filtering can change the values available for later mapping and the resulting color range. See statistical ROI rules.

For P Value, the reader uses the NIfTI description from SPM, REST or DPABI (`SPM{T_[20]}`, `REST{F_[3,20]}`, `DPABI{R_[20]}`, or a Z token); a standard NIfTI CORREL/TTEST/FTEST/ZSCORE intent and its parameters are a fallback. Missing or invalid degrees of freedom cause an error, never a silent numeric-threshold fallback. T, R and Z use **two-sided** uncorrected P; F uses the **upper tail** and excludes negative F values. The resulting statistic cutoff, degrees of freedom, tail and metadata source are recorded in render preparation/provenance. P filtering occurs **before** mapping; interpolation or neighborhood smoothing can subsequently change vertex values, so a displayed vertex value is not itself a voxelwise P-value test. There is no family-wise/FDR correction or spatial registration.

Mapping algorithm (Surface Map)

No. / menu label	Calculation and boundary caution
1 · Nearest Voxel	Round the transformed surface position and sample that voxel. Invalid/out-of-volume positions receive zero.
2 · Average Surface Neighbors	Take algorithm 1 values first, then average each vertex with its surface-mesh neighbors and itself. This is a surface-adjacency average, **not** a 3×3×3 voxel mean; mesh topology affects the result.
3 · Average 3-by-3 Voxel Neighborhood	Apply a 3×3×3 mean filter to the volume, then nearest-voxel sample it. Volume boundary handling follows the filter implementation.
4 · Gaussian Voxel Smoothing	Apply MATLAB `smooth3(...,'gaussian')` to the volume, then nearest-voxel sample it. This alters sampled values and may mix nearby labels/signs.
5 · Trilinear Interpolation	Interpolate the eight enclosing voxel values at an interior position; outside the supported interior returns zero.
6 · Maximum 3-by-3 Voxel Neighborhood	At a valid interior rounded position, take the maximum of 27 voxels.
7 · Minimum 3-by-3 Voxel Neighborhood	At a valid interior rounded position, take the minimum of 27 voxels.
8 · Extremum 3-by-3 Voxel Neighborhood	Choose the 27-voxel sample with greatest absolute magnitude and **retain its original sign**.

The mapping gallery illustrates how the eight algorithms display the same input. It is not an accuracy ranking. For a discrete atlas, start with Nearest Voxel: averaging and interpolation can create values that are not source labels. For continuous statistics, algorithms 2 and 5 can give smoother displays; they respectively average sampled mesh neighbors and interpolate the volume at each vertex. Record the chosen method and compare the result with the source map before interpreting a location.

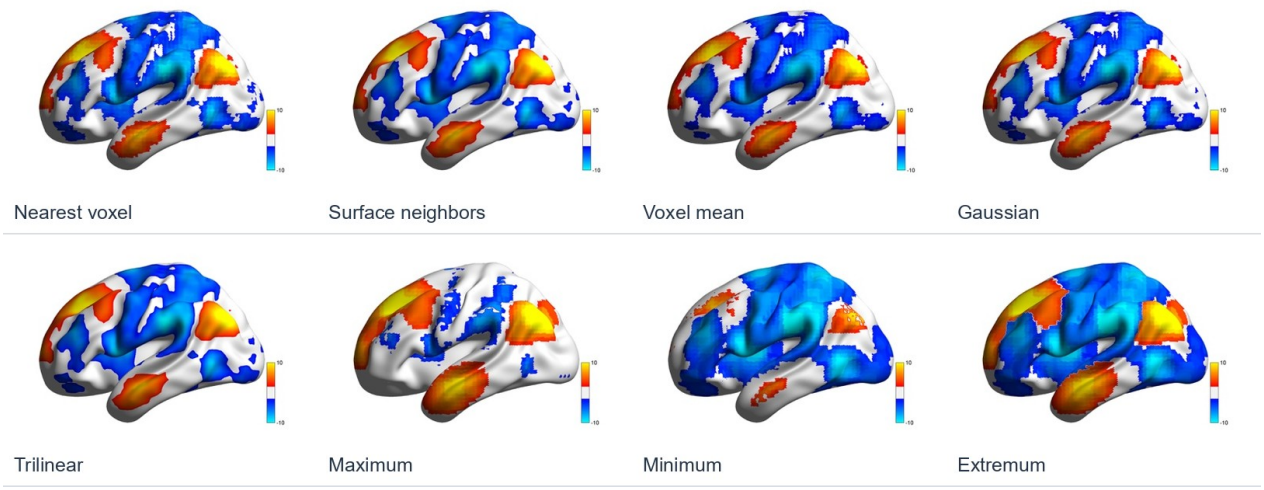


Figure 15. Eight volume-to-surface mapping algorithms applied to the same input.

For sign and color-range choices, **Display Values** changes the signed color axis but does not remove negative or positive vertex values. If a map contains values from −3 to +6, Both mode uses a symmetric −6 to +6 axis; a +3 and a −3 therefore have comparable distance from the center even though +6 is the observed positive maximum. Positive Only or Negative Only uses the corresponding half of the palette. Change the relevant bounds only after checking the actual sampled value distribution and the displayed colorbar; do not interpret white/null color as a formal nonsignificant voxel unless the thresholding and adjustment rules warrant that interpretation.

CPU supports all eight. The current GPU mapping path supports algorithms 1 and 5 only; an explicit GPU request for the other algorithms falls back to CPU with a recorded reason. Auto chooses GPU for algorithm 5 on sufficiently large surfaces when a supported GPU is available; otherwise it uses CPU. This is an execution policy, not a guarantee of speedup on every dataset.

Ranges and sign display

Control	New-settings default	Current rule
Positive Range / Negative Range	Auto / Auto	Enter `Auto` or minimum and maximum (either endpoint may be Auto). Manual bounds persist when the volume file changes and are identified in Scene Summary; recheck them for each new dataset. **Auto Ranges** only resets the pending fields—click **Apply** to commit and render.
Colormap	Blue-White-Red	One 256-entry palette: its lower half represents negative values and upper half positive values. A custom RGB palette can be set with Colors.... The Surface panel's scalar-map palette is separate.
Display Values	Positive & Negative	Both, Positive Only or Negative Only changes the **color axis, not which vertex values exist**. Both uses a symmetric `[-M, +M]` axis around zero even when the data extrema differ; one-sided modes use the respective palette half. Values outside a chosen range clamp to endpoint colors.
Adjust for Null / Null Color	on / `[0.95 0.95 0.95]`	When on, the null-side palette region receives Null Color; when off, finite values use the continuous palette. Null Color is editable only when adjustment is on. This changes display, not sampled values.

Both-mode symmetry can reduce contrast for the smaller-magnitude sign. Do not infer absence of an effect merely because one side is pale; inspect the numeric data and actual colorbar. The illustrated example uses a resting-state example map from a published study, but its alignment to a chosen surface still needs independent verification. See signed-axis details.

Multiple-comparison correction is **not available**. Check the source map's statistic, degrees of freedom, coordinate alignment and your analysis-specific inference policy before interpretation.

10 ROI drawing

Set **Display Volume** on and choose **Volume Type** → **ROIs (Atlas / Statistical)** to draw atlas or statistical regions. **Apply** commits and renders; **Revert** discards pending edits. ROI identifiers are labels, not continuous quantitative values.

The shared threshold controls are above **Surface Mapping** in the Volume panel, not inside that surface-only section. In ROI mode, P Value and Cluster Size are enabled together; Statistical Threshold is used by Value/Cluster instead.

ROI controls

Control	New-settings default	Current rule
Threshold Mode	None	None preserves positive integer atlas IDs. Value and Cluster construct connected regions with `abs(value) >= abs(Statistical Threshold)`; Cluster also removes components smaller than Cluster Size. P Value converts supported NIfTI T/F/R/Z metadata into a statistic cutoff, then applies Cluster Size before assigning ROI IDs.
Statistical Threshold / P Value / Cluster Size / Voxel Connectivity	0 / 0.05 / 0 / 18 neighbors	Value/Cluster use the numeric threshold; P Value requires `0 < P < 1` and valid statistic metadata. T/R/Z are two-sided; F is upper-tail only. P Value and Cluster use Cluster Size in voxels, not mm³. Zero and nonfinite voxels are excluded; choose 6, 18 or 26 connectivity.
Draw All ROIs	on	Draws every valid positive ID. While on, **ROI Selection** is disabled.
ROI Selection	empty	Turn Draw All off to enter positive IDs separated by spaces or commas. Typing a nonempty list also turns Draw All off. Atlas mode uses source IDs; statistical Value/Cluster/P Value modes use newly assigned, consecutive component IDs, which can change with thresholds, Cluster Size or connectivity.
ROI Color Mode	Label Index	Uniform uses one ROI Color; Label Index cycles the selected categorical ROI Palette by ID; Custom requires an explicit `ROI_ID R G B` table. Color modes do not redefine region membership.
ROI Palette	HSV	Label Index can choose any named palette, including Viridis, Plasma, Inferno, Magma and Cividis. Its 256 colors cycle by positive ROI ID; this is categorical indexing, not a continuous quantitative colorbar. HSV and Lines generally separate categories better than sequential gradients.
Show ROI ID Legend	off	Optional ROI ID/color key in the first rendered view; at most 16 entries are shown, with an overflow count. It is not a value colorbar.
ROI Color / Load ROI Colors... / Custom table	`[0.8 0.2 0.2]` / no table	ROI Color applies in Uniform. For Custom, enter or load rows `ROI_ID R G B`, unique positive integer IDs and finite RGB in [0,1]. Every selected ID present in the volume needs a color; missing rows cause an error rather than a fallback.
ROI Smoothing / Smoothing Kernel	Gaussian / 3	None, Box or Gaussian. Kernel is in voxels; 1 disables smoothing, and larger effective odd kernels soften boundaries. This is display mesh smoothing, not a change to the stored volume or scientific cluster IDs.

ROI P Value follows the legacy sequence: convert P to a statistic cutoff, discard subthreshold/nonfinite voxels, remove components smaller than Cluster Size, then number surviving components. Positive and negative T/R/Z voxels in contact can form one ROI; F excludes negative values. The cutoff, degrees of freedom, metadata source, removed clusters and resulting IDs are recorded in preparation provenance. Missing/contradictory metadata raises an error. This is an **uncorrected voxelwise threshold**; Cluster Size does not by itself establish cluster-level significance, and no multiple-comparison correction or registration is performed. An atlas should normally use None, not P Value.

ROI meshes use a 0.5 isosurface and a virtual zero halo to close regions touching volume borders. Smoothing checks each original 26-connected component and conservatively falls back to the whole original ROI if one would disappear. This does **not** guarantee volume, thin-branch or full topology preservation. Active smoothing with kernel >1 requires Image Processing Toolbox (`bwconncomp`); without it choose None or kernel 1. See ROI geometry, custom colors and statistical numbering. Recheck a Custom table whenever statistical filtering changes IDs. Automatic atlas/statistic classification is not provided.

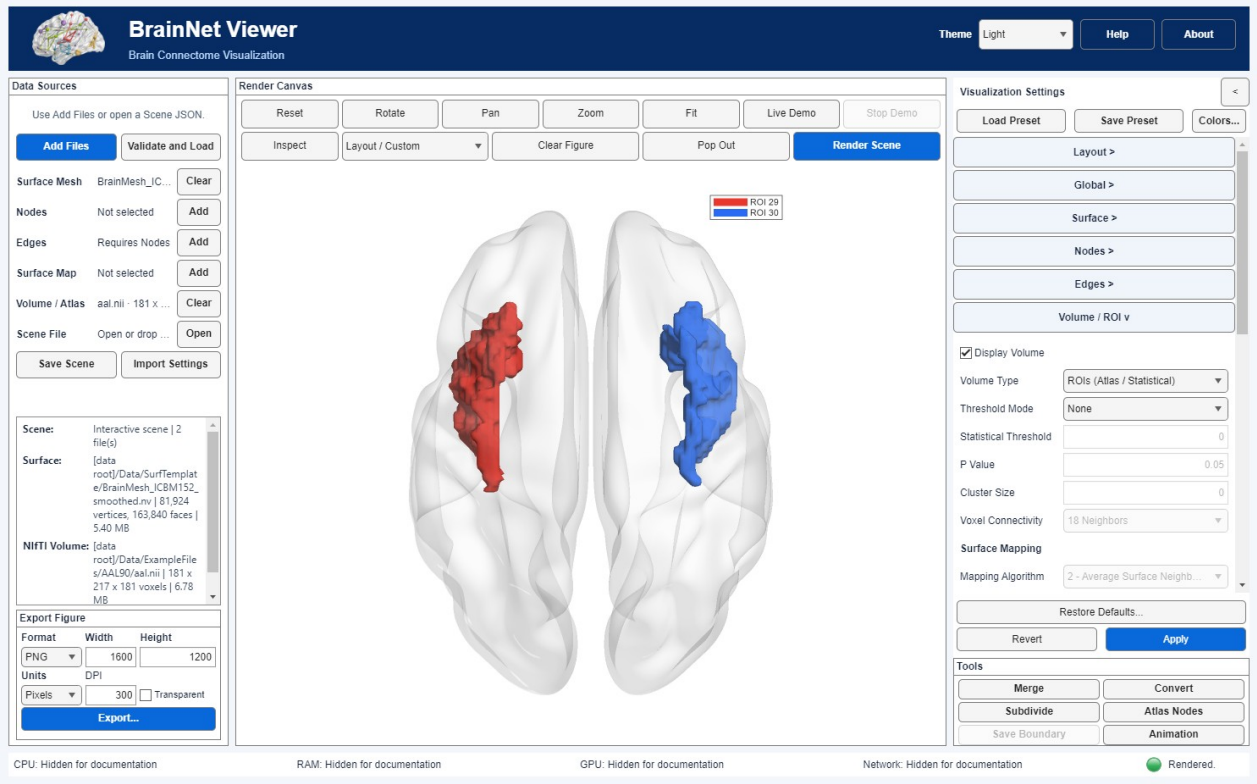


Figure 16. Bilateral AAL insula regions displayed above a translucent surface.

11 Colormaps and presets

Use custom colormaps to control display colors and visualization presets to reuse settings across datasets. A colormap changes appearance, not numerical values, thresholds or registration. A preset transfers display settings, not data files.

Custom RGB colormaps

Open **Colors...** below **Load Preset / Save Preset** in the right-hand Visualization Settings panel. The editor has four independent **Target** choices: **Surface**, **Nodes**, **Edges** and **Volume**. Volume uses one signed colormap, not separate positive and negative palettes. Named choices now include Viridis, Plasma, Inferno, Magma and Cividis; these are also available directly from each target's colormap selector. Choose a named **Preset** and **Count** (2–4096 rows), then **Generate**; **Resample** interpolates the current table to Count, and **Reverse** reverses its row order. The Red/Green/Blue table accepts finite components in [0,1]; a valid edit updates the preview. **Import...** reads RGB TXT/JSON; **Export...** writes the current target's table only, without applying it. Existing export files require an overwrite decision.

Use in Settings stages the four maps in the right-hand panel and closes the editor. **Cancel**/closing discards editor changes. A merely viewed named map stays named; modifying it produces an embedded numeric map. Choose a display mode that consumes that target's colormap, then press the single panel-bottom **Apply**: it commits settings **and renders**. **Revert** discards pending panel edits. A separate Render Scene click is unnecessary after a successful Apply.

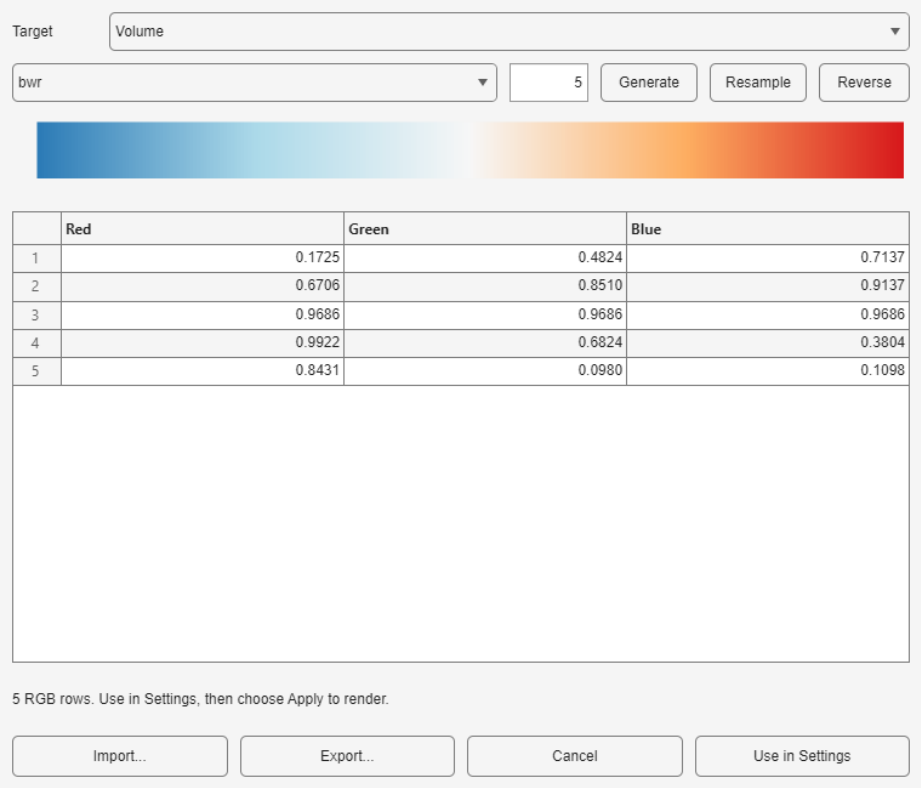


Figure 17. The custom Volume colormap editor before the colors are applied to the scene.

Surface Scalar, Node Mapped/Modules, Edge Mapped, and Volume Surface Map are examples of applicable consumers. In **Edge Node Module** mode, same-module edges inherit the **Node** module palette, not the Edge colormap. Node and Edge fixed threshold colors do not use the continuous palette. For **Volume Display Values = Positive & Negative**, the one map is split at its midpoint between negative and positive values; review the resulting signed colorbar and limits. The ROI **Label Index** palette is a separate categorical selector in Volume settings, not an editor target. Numeric palettes are RGB-interpolated, not guaranteed perceptually uniform or accessible. See the node, edge, volume and ROI chapters before interpreting colors scientifically.

RGB TXT accepts three numeric channels per row; expressions are not executed. JSON uses `kind: brainnet-colormap`, schema `1.0`, and a `colors` array. File import is limited to 1 MiB. Scene and preset JSON retain applied custom arrays; the editor's Export... creates only a portable colormap file. The legacy renderer does not silently substitute unsupported custom arrays. Treat imported legacy MAT configuration files as trusted-code inputs and review old combined volume palettes manually.

Save and load a display preset

1. Load a dataset and edit Visualization Settings. Press **Apply** to commit and render.
2. Press **Save Preset** to write `*.preset.json`. This captures **applied** settings, not uncommitted edits, input paths, camera, theme, compute backend or export-panel preferences.
3. With another dataset, press **Load Preset**. Read the compatibility review and confirm or cancel. Confirmation replaces all nine display sections and pending edits, retains the current input references and export preferences, and clears the old render/camera. Press **Render Scene** after loading.
4. Recheck data-specific color limits, thresholds, units, selected ROI IDs and anatomy before exporting. A role-compatibility warning is not proof of spatial registration or scientific validity.

Save Scene is different: it records one Scene's input references, settings and (when saved from the modern App) session camera/export preferences; input files are referenced, not bundled. Invalid, unknown-version or explicitly input/camera-bearing preset JSON is rejected. Cancellation and failed validation do not replace the current settings. Fiber rendering remains unavailable; Surface Map and statistical ROI P-value presets require supported statistic metadata in the selected NIfTI. Preset names/titles may contain private text; review before sharing.

The real-data illustrated exercise shows custom Volume RGB staging, Apply and preset replay. Check the resulting colorbar and limits against the loaded data before reusing a preset in a figure.

12 Figure and movie export

Export requires a currently rendered Scene; a loaded but unrendered Scene cannot be exported. Export is an output operation, not a change to the scientific data or applied settings.

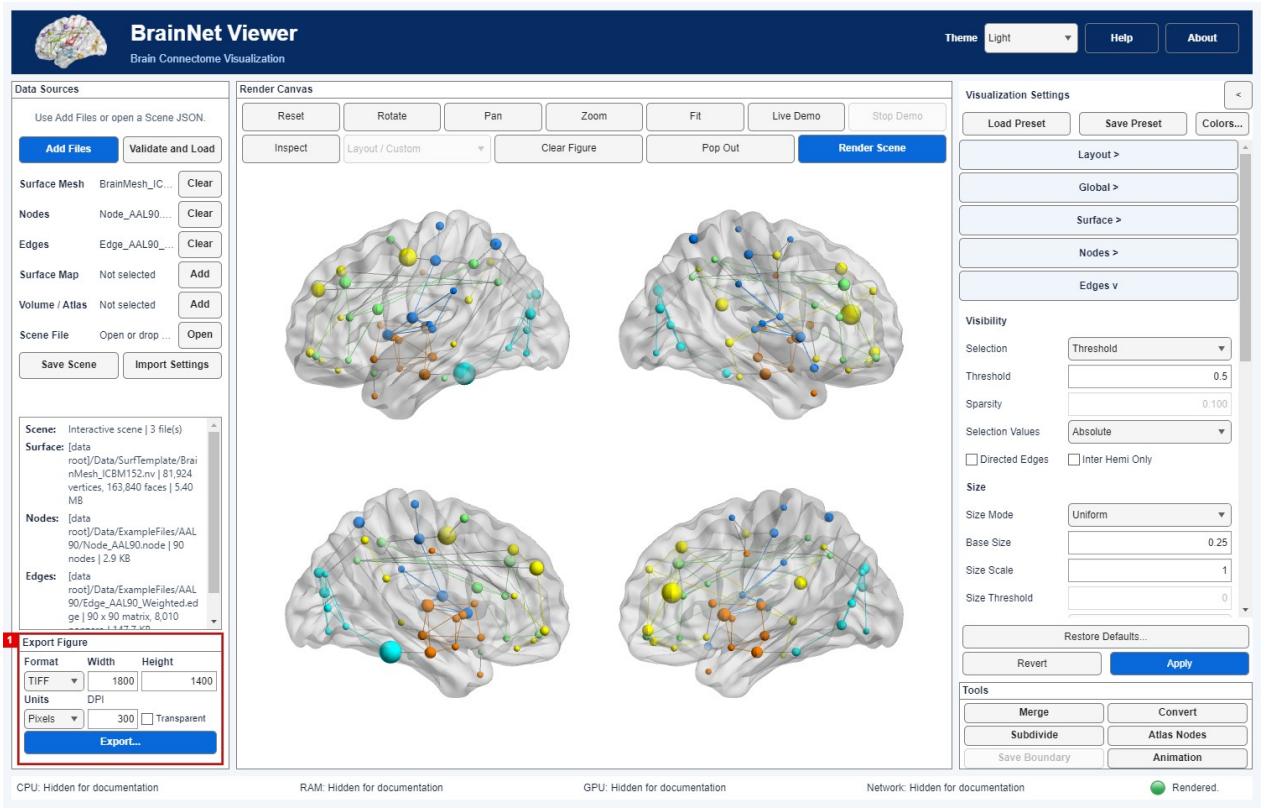


Figure 18. The red frame identifies Export Figure settings, including output format, dimensions and resolution.

Export a still figure

After loading, configuring and rendering, use **Export Figure** below Data Sources. Select the format and size, then click the button and choose a file. The current panel defaults are **PNG**, **300 DPI**, **1600 × 1200**, **Pixels**, **Automatic** content and **White** background. These are export-panel defaults, distinct from the `image` settings section in a visualization preset.

Control	Current choices and meaning
Format	PNG, JPEG, TIFF, BMP, PDF or SVG. Raster formats produce pixels; PDF/SVG do not guarantee editable vector 3-D content.
DPI, **Width**, **Height**, **Units**	Positive DPI and dimensions; units are Pixels, Inches, Centimeters or Points. Set the target's intended physical size and resolution before saving; inspect the actual output dimensions.
Content	Automatic or Image in the current desktop panel. Image forces raster content; Automatic may rasterize complex 3-D graphics. Do not assume a PDF/SVG is vector-only.
Background	White, Black or Transparent (Experimental). JPEG/BMP cannot be transparent. On supported formats, unsupported/missing alpha is reported rather than silently replaced with white; test the result in the intended downstream software.

The file chooser may replace the selected output; choose a new path if an existing file must be preserved. A successful export writes the figure **and** a neighboring `.provenance.json` sidecar with Scene/renderer/export information. Review that JSON and any absolute input paths or titles before sharing. Output preferences are part of an App-saved Scene session, but loading a display preset does not change them. The interface Light/Dark theme is not the figure background.

Export a camera animation

Render first and adjust the canvas camera, then choose **Export Animation** in Tools. The animation dialog exports the current static scene into a **new folder**; it does not animate changing connectivity or edit the main canvas/camera. Its defaults are **960 × 720 pixels**, **5 seconds**, **24 fps**, world **Z** axis, **0°→360°**, and **Safe Fit on**. Available formats are MP4 when MATLAB's MPEG-4 profile exists, Motion JPEG AVI, and PNG frame sequence. Choose **Export...** and a new folder name under an existing parent; existing bundle paths are never overwritten.

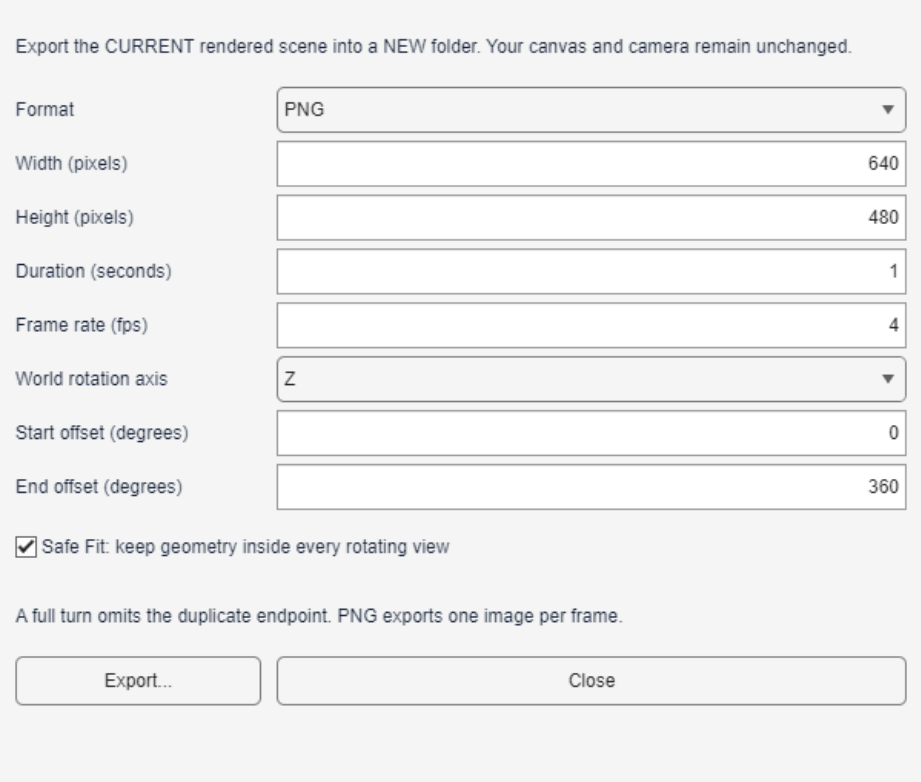


Figure 19. Camera-animation controls for a PNG frame-sequence example.

Safe Fit centers and orthographically frames geometry across the orbit on a private rendering copy; disable it only if preserving current framing matters more than possible cropping. Angles are offsets from the current camera around the chosen world X/Y/Z axis. A full turn omits its duplicated endpoint. Frame count is `round(duration × fps)`, so effective duration may differ slightly. Export speed is not playback speed; no GPU video-encoder claim is made. The dialog's **Cancel Export** takes effect after the current frame/codec operation; partial output is cleaned up.

The new folder contains `movie.mp4` or `movie.avi`, or `frame-000001.png` etc., plus `preview.png` and `provenance.json`. The latter can contain Scene references, input metadata, cameras and per-frame records; source inputs are **not** copied. Limits: 128–3840 pixels per dimension, 1–60 fps, 2–1800 frames, at most 120 requested seconds, one billion total frame-pixels, and a nonzero orbit no greater than 360°. Video dimensions must be even; there is no transparent movie or audio. Codec availability and pixel-identical results across platforms are not guaranteed. See the animation guide for API and verification details.

13 Surface formats and mesh tools

A successfully read mesh is **not** proof of MNI registration, matching units, compatible handedness, or permission to redistribute it. The modern readers preserve raw geometry rather than silently applying a dataset-specific transform.

Load and review a surface

In **Data Sources** → **Surface Mesh**, select a supported mesh; then **Validate and Load**, configure the display, and **Apply** to render. The same independent readers serve the desktop, Scene JSON and `BrainNetRender`. Surface support includes `.nv`, FreeSurfer `.pial`, BrainVISA `.mesh`, BYU/MOVIE.BYU `.g`, MNI/BIC `.obj`, MZ3 `.mz3/.mz3.gz`, and GIfTI `.gii/.gii.gz`, within each reader's restrictions. `.obj` means the MNI/BIC polygon format here, **not** generic Wavefront OBJ. `.mesh` does not imply Medit format. `.pial` is the advertised FreeSurfer desktop suffix; do not assume arbitrary FreeSurfer suffixes or compressed `.pial.gz` work. See the FreeSurfer, BrainVISA, BYU, MNI/BIC and MZ3 guides for exact encodings and rejection rules.

Coordinates and face winding are kept as returned by each reader; disk indices become one-based internally. BrainVISA `.mesh` **does not** receive the original BrainNet's implicit XYZ inversion. FreeSurfer surface RAS is not automatically scanner RAS/MNI. GIfTI transform metadata is recorded, not automatically applied by the mesh utilities. MZ3 embedded RGBA/scalars are validated but not automatically promoted to display settings or Surface Map. Verify space and geometry alignment before combining a mesh with nodes, volume, atlases or other surfaces. A single unsplittable mesh should use applicable full-surface layouts; two-hemisphere layouts cannot be assumed from its filename.

Tools that write new data

Tool in Tools	Inputs and operation	Output / caution
Merge Surfaces	Automatically review 2–64 meshes, order, bounds, counts and SHA-256. Space, units and note are optional; no verification checkbox is required.	Concatenates coordinates/faces into a new ASCII `.nv`; does not register, weld, smooth, repair or transfer color/scalar attributes.
Convert Surface to NV	Review one supported mesh and optional space/unit declarations.	New `.nv` preserving reader-returned geometry; no implicit legacy `.mesh` coordinate inversion or attribute transfer.
Subdivide Surface to NV	Review one mesh; choose 1–3 linear midpoint levels (4× faces per level).	New denser `.nv`; no smoothing, curvature preservation or automatic node/map correspondence.
Generate Nodes from Atlas	Review a discrete integer NIfTI label atlas, selected region IDs, affine and declared space/units.	New `.node`; each region uses its unweighted voxel-centre centroid, which need not lie within a concave/disconnected region. No registration or name lookup.

Each tool has a review step, checks source/dependency fingerprints again before publication, writes a **new** destination without replacing an existing one, and leaves the active Scene unchanged. Mesh utilities use default combined limits of five million vertices and ten million triangles; read/allocation may still require substantial RAM. The `.nv`/`.node` output includes a `#` JSON provenance comment, potentially with absolute source paths. Do not publish it without inspecting privacy and licensing. Declarations in comments do **not** assign a Scene coordinate space or perform transforms. Load the new file explicitly through Data Sources; for a portable Scene, place it within the configured data root or use an appropriate root. See merge, convert, subdivide, and atlas nodes for exact review fields, formats and limits.

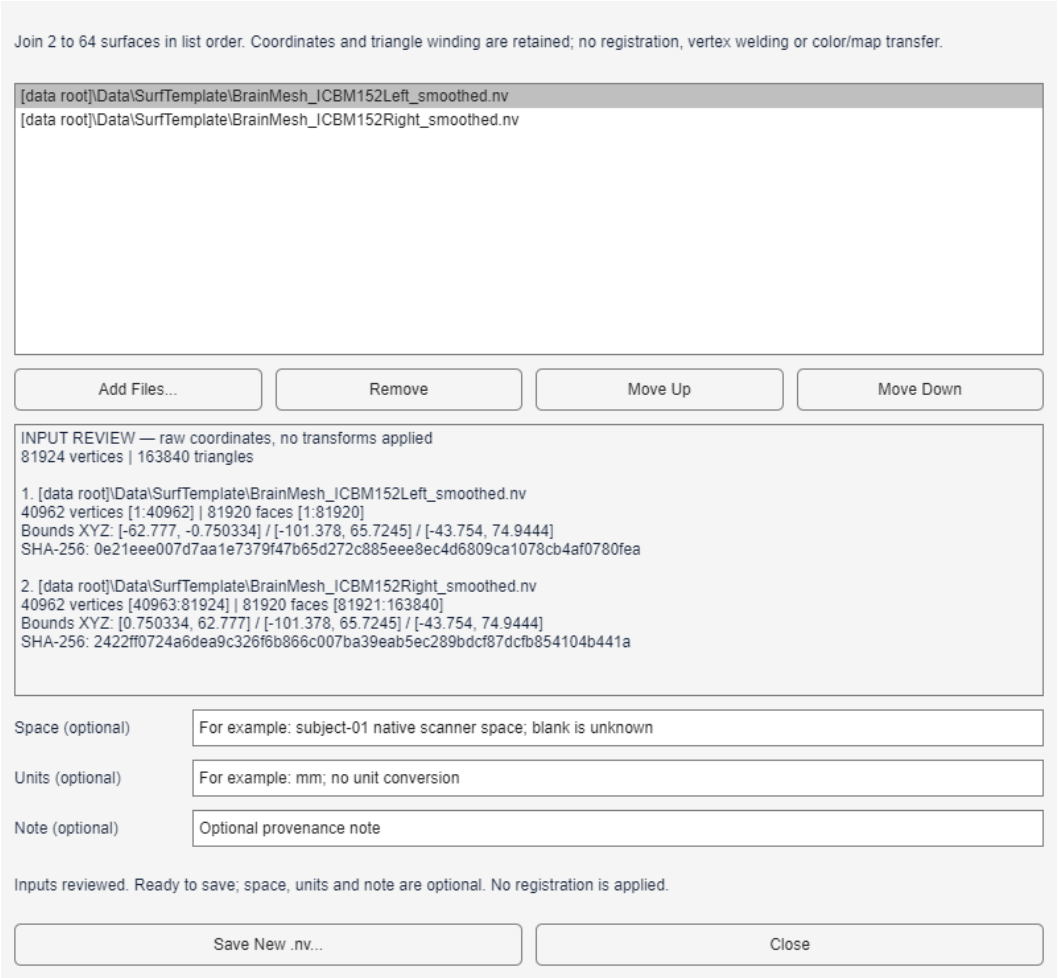


Figure 20. Merge Surfaces automatically reviews the selected inputs. Space, units and note are optional.

These geometry operations are not a scientifically validated surface-registration pipeline. Verify anatomy and resource permissions for each dataset before using or distributing derived files.

14 Command-line rendering and provenance

Command-line rendering uses the **modern** Scene/renderer without opening the desktop App; it does not replace the historical `BrainNet_MapCfg` signature. A MATLAB license and working graphics environment are still required. This is not a browser/Docker edition.

Prepare a reproducible Scene

In the desktop, select files, **Validate and Load**, adjust Visualization Settings, and **Apply** (which commits and renders). Use **Save Scene** to create Scene JSON. It records applied settings, relative input references/hashes, coordinate/redistribution metadata, and—when saved by the modern App—optional camera and export-panel state. It does **not** embed source data, accept unapplied edits, certify spatial registration or grant file-sharing rights. The data root is an explicit directory: references resolve relative to that root, not relative to the Scene JSON's folder. If external files are selected in the App, approval stages verified copies under its configured root; a manually written Scene or CLI run does not silently copy outside data.

To render one image without opening the App:

```
record = BrainNetRender("D:/results/scene.json", "D:/results/figure.png", ...
    DataRoot="D:/project-data", ComputeBackend="cpu", ...
    PerformanceMode="quality", Width=2000, Height=1500, ...
    Resolution=300, Overwrite=false);
```

The command reads and validates input files, prepares data, renders a hidden modern figure and writes the image plus `figure.png.provenance.json`. Output extensions are PNG, JPEG, TIFF, BMP, PDF or SVG; complex PDF/SVG scenes may contain rasterized 3-D content. The output file and sidecar are protected from overwrite by default. Use a distinct output name for every run or explicitly choose `Overwrite=true` after checking the targets. A simple MATLAB loop may call `BrainNetRender` over different Scene/output paths; there is no claim of an independent job scheduler or automatic dataset discovery.

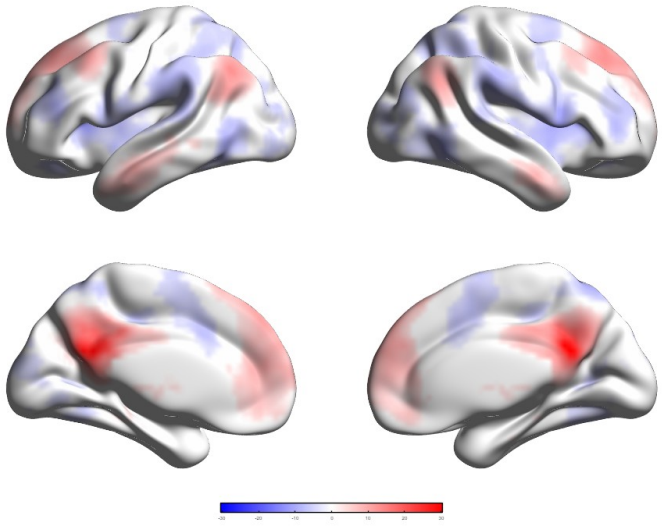


Figure 21. A four-view figure rendered by BrainNetRender from a real-data Scene JSON.

Options to review per run

Option	Current behavior
`DataRoot`	Defaults to the installed BrainNetViewer folder; set it explicitly for portable project data.
`ComputeBackend`	`auto` (default), `cpu`, `gpu`; GPU applies only where individual operations support it.
`PerformanceMode`	`quality` (default), `interactive`, `auto`. For publication runs, explicitly request quality.
`Width`, `Height`, `Resolution`, `Units`	Specify both dimensions or neither. Otherwise saved App export preferences are used, falling back to Scene `image` defaults. Explicit dimensions without Units use pixels. DPI and physical units are interpreted by the exporter.
`BackgroundColor`, `ContentType`	Explicit choices override saved preferences. JPEG/BMP cannot use transparent `none`; raster formats reject vector content. Auto/PDF/SVG do not promise pure vectors.
`LegacyConfig`	Optional trusted legacy EC MAT conversion before rendering; not a full old-GUI replay or a sandbox for untrusted MAT objects.
`Planner`	Optional reusable `brainnet.render.RenderPlanner` object; within one MATLAB process it can reuse preparation work when keys match. It does not prove GPU speedup or identical pixels.

Scene loading checks input identities/hashes; a changed or missing file should stop instead of silently substituting data. The Scene's optional camera is restored only when its content signature remains compatible; otherwise the layout default is used. For comparisons across machines/versions, review settings, numeric fingerprints, renderer/version and camera records in the provenance JSON; do not require pixel-identical image hashes across graphics drivers. **The sidecar may contain local paths, titles and sensitive metadata**. Review it and input-resource licenses before publication. Nothing is uploaded automatically. For reusable appearance without input paths, use a visualization preset rather than Scene JSON; see Chapter 11.

The real-data CLI tutorial demonstrates one portable Scene and two output formats. Review sidecar privacy and cross-platform rendering differences for your own project; this chapter is not a Docker deployment guide.

15 Migration and troubleshooting

BrainNet Viewer provides a MATLAB desktop workflow. Standalone Runtime-only and Docker/browser editions are not included. This chapter explains supported configuration formats, interaction controls and troubleshooting.

Moving from BrainNet Viewer 1.7/1.8

Older workflow	Current route	What to check
GUIDE main window / Load File	`BrainNetViewer`; Data Sources → Validate and Load	Use the project launcher and an explicit data root. The App asks before staging an outside-root file; Scene JSON does not bundle its input files.
BrainNet_Option.fig and OK/Apply	Right-hand Visualization Settings → shared **Apply**	Apply commits **and renders**. **Revert** drops pending edits; canvas **Reset** changes only the camera; **Restore Defaults** replaces display settings after confirmation.
Save/Load Option MAT	**Save/Load Preset** for display reuse; **Import Settings** for trusted EC MAT	Presets omit inputs/camera/export-panel preferences. An EC import is a reviewed conversion, not guaranteed lossless reproduction; retain the original MAT and inspect `fields_requiring_review`.
Save or reopen an analysis	**Save Scene / Open Scene JSON**	Scene records input references/hashes and applied settings, with optional modern camera/export state. Keep the corresponding data-root structure; verify hashes and spatial alignment.
Redraw / Clear Figure	**Render Scene** for explicit redraw; **Clear Figure** for view-only clearing	Clearing the figure preserves inputs/settings; a per-role Clear removes that input.
Get position	In-canvas **Inspect**	Queries AAL/Brodmann using assumed MNI-like coordinates; it does **not** register or transform the input. Verify provenance before citing regions.
Merge Mesh / generate coordinates	**Merge Surfaces**, **Convert Surface to NV**, **Subdivide Surface to NV**, **Generate Nodes from Atlas**	New reviewed files, no silent source overwrite or registration. Conversion is geometry-only; the old `.mesh` implicit coordinate inversion is not retained.
Movie / print / script	**Export Animation**, **Export Figure**, `BrainNetRender`	Animation is a static-scene camera orbit. Print from an exported image/PDF externally; the modern CLI is not a drop-in `BrainNet_MapCfg` replacement.

Locate familiar menus and interaction tools

The desktop follows one sequence: load files, configure, render, interact and export. Data Sources selects and validates the inputs; Visualization Settings controls appearance; the central canvas displays the result. Help and About are in the header. Separate tool windows provide mesh and atlas-node operations.

After rendering, use the canvas **Rotate**, **Pan** and **Zoom** tools to adjust individual views. **Fit** frames the visible geometry; **Reset** returns to the layout's initial camera. **View Direction** supplies Left, Superior, Anterior and opposite/other directions for Single View; select the desired direction and render again. **Inspect** replaces the old Get Position datatip workflow and clears its compact on-canvas label when clicked again. **Live Demo** and **Stop Demo** control an on-screen camera orbit, while **Export Animation** makes a separately configured movie or PNG frame sequence.

Native printing and MATLAB Plot Edit are not provided. Export a figure at the required pixel/physical size, then print or annotate it in a suitable external application. For saving configuration, choose a visualization preset for reusable appearance or a Scene JSON for reproducible input references and applied settings. The historical `BrainNet_MapCfg` file remains a legacy MATLAB entry point in this development tree, but `BrainNetRender` is the documented modern Scene-based command; do not assume interchangeable argument or provenance behavior.

Modern Settings schema is 1.22.0. Older Scene/preset/EC values may migrate, but their appearance can change because node/edge threshold endpoints, modules, signed volume color handling, ROI numbering/smoothing and other semantics were clarified or repaired. Recheck science-critical settings with a small dataset that independently varies `.node` columns 4 and 5 and includes values exactly at thresholds. Do not use visual similarity alone as proof of parity. For source-specific rules see Nodes, Edges, Volume, ROI/Inspect and Colormaps.

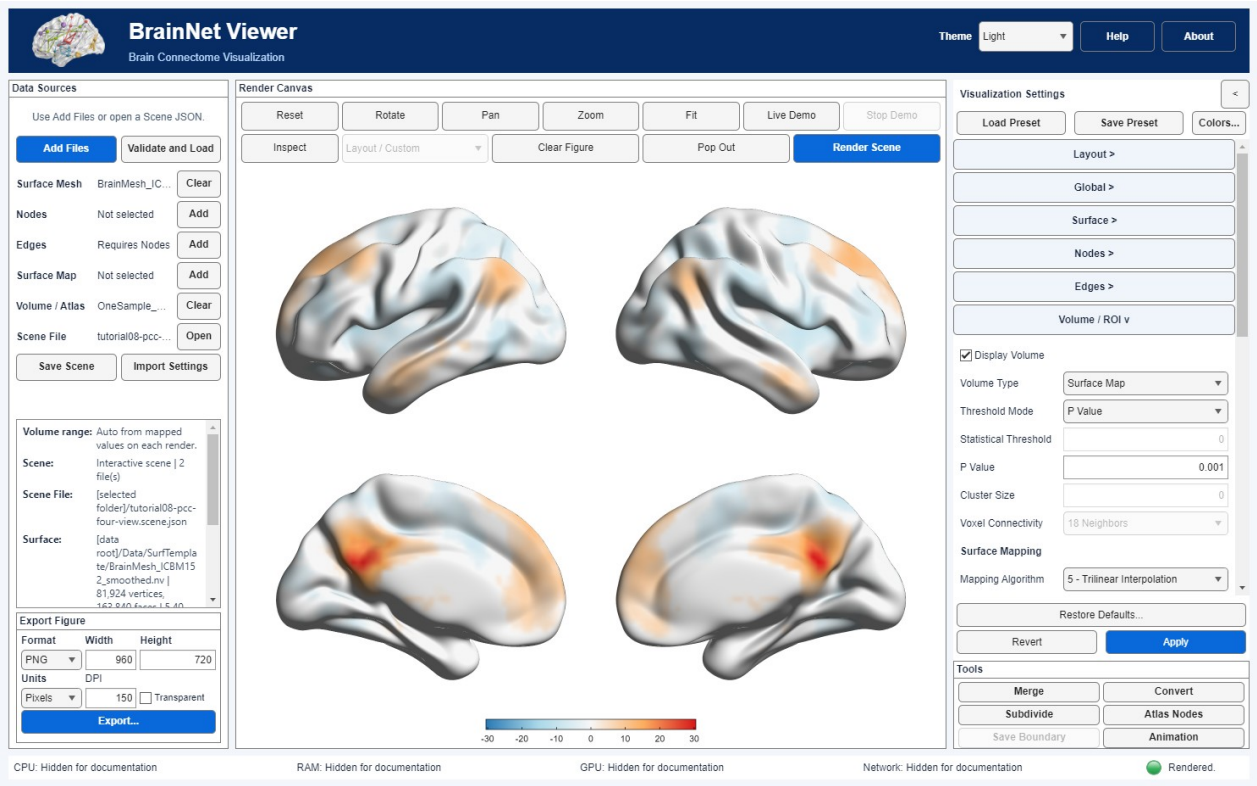


Figure 22. The current main window brings Data Sources, render controls, Visualization Settings and export into one workspace.

Current boundaries, not silent promises

- The independent NIfTI reader supports NIfTI-1/2 single files and paired headers/images, including supported gzip variants; **pure Analyze 7.5** without NIfTI magic is rejected. MNI/BIC `.obj` is not Wavefront OBJ. FreeSurfer `.pial`, BrainVISA `.mesh`, MZ3 and GIfTI have narrower reader-specific rules. Loading a geometry does not establish coordinate registration.
- ROI **P Value** now converts supported statistical NIfTI metadata but does not infer whether an input is an atlas or statistic map, correct multiple comparisons or prove cluster-level significance. Fiber-tract rendering, arbitrary **Plot Edit**, GIF/EPS/EMF export, native print and Runtime-only packaging are not available/validated in this build. A proposed Python/Desktop or Docker version is future work, not a current feature.

- The current desktop target is Windows MATLAB R2025a Update 1. Other MATLAB versions, platforms, all graphics drivers/monitor DPI combinations and required-toolbox matrices are not fully certified. The startup build date is development metadata, not the release date. Materials used internally may lack redistribution clearance.
- GPU acceleration is operation-specific, not a promise that all rendering or video encoding runs on GPU. Complex PDF/SVG scenes may be rasterized; JPEG/BMP have no transparent background. PNG/TIFF transparency is conditional on an actual alpha-bearing export. Hashes and provenance identify data/configuration, not anatomical validity, publication rights or pixel identity across systems.

If something goes wrong

Symptom	First check / safe action
The launcher is not found or a different BrainNet starts	Use the current project root containing `BrainNetViewer.m`, add that root to the MATLAB path if necessary, and remove conflicting path entries from the session; do not alter the original software folder. See getting started.
External input cannot become a portable Scene	Respond to the App's **Stage and Add** prompt, or deliberately launch with an appropriate data root. Do not hand-edit absolute paths into portable Scene references. Review copied companions and permissions.
Scene/preset JSON fails to open, or files/hashes do not match	Use **Open Scene JSON** for Scene files and **Load Preset** for `*.preset.json`; these schemas are not interchangeable. Check version/required fields, data-root-relative paths, companion files and the original hashes. Do not suppress validation by changing a hash.
New file selected but the old image remains / nothing exports	**Validate and Load**, then **Apply** (commits and renders) or explicit **Render Scene**. Export needs a current Modern Batch render. Loading a preset clears the old render, so Render Scene again. Check the Scene summary for retained manual volume limits when changing datasets.
Wrong side, scale, or atlas region / empty split view	Check the actual coordinate frame, orientation, units, surface-volume alignment and hemisphere separability. The App does not register data. Use a supported whole-surface layout for an inseparable mesh; treat Inspect's AAL/Brodmann result as provisional.
ROI P Value metadata error, unsupported file, or custom color error	Verify NIfTI statistic type/degrees of freedom; do not treat an atlas as a statistic map. Otherwise choose a compatible reader/file or provide every selected ROI ID in the custom color table; see ROI rules.
GPU, transparency or animation export fails	Try the documented CPU backend where relevant; verify output format, real alpha support, video codec availability, even video dimensions, frame/size limits and a **new** animation folder. Do not overwrite or publish partial output.

For an unresolved reproducible failure, record the build/MATLAB version, data roles and non-sensitive dimensions, exact error, minimal steps, compute backend and output format. Keep participant data and absolute provenance paths out of public reports unless authorized. Verify limitations against the actual package before sharing results.

Appendix References and acknowledgments

The references below identify the principal software and atlas sources. Cite the actual atlas release, derivative and analysis method used in your study, rather than treating this general list as a substitute for dataset-specific attribution. The 2013 article is the primary software reference for BrainNet Viewer.

Software and atlas references

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